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MDS Diffusion Layers for Arithmetization-Oriented Symmetric Ciphers: The Rotational-Add Construction

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Abstract. We introduce the rotational-add diffusion layers aimed for applications in the design of arithmetization-oriented (AO) symmetric ciphers, such as fully homomorphic encryption (FHE)-friendly symmetric ciphers. This generalizes the rotational-XOR diffusion layers which have been utilized in the design of many important conventional symmetric ciphers like SHA-256, SM4, ZUC and Ascon. A rotational-add diffusion layer is defined over the finite field $\mathbb{F}_p$ for arbitrary prime p, enabling implementations using only rotations and modular additions/subtractions. The advantage of using such diffusion layers in AO ciphers is that, the costs of scalar multiplications can be reduced since the appearing scalars include only ± 1, thus the total costs depend on sizes of the rotation offsets. In this paper, we investigate characterizations and constructions of lightest rotational-add diffusion layers over $(\mathbb{F}_p^m)^n$ that are maximum distance separable (MDS) with a focus on the case $n = 4$. It turns out that the minimum achievable size of the rotation offsets is 5 subject to the MDS property constraint. We specify a large class of rotational-add diffusion layers with 5 rotations and traverse all possible patterns of appearance of the scalars ± 1. In four cases we can derive computationally tractable necessary and sufficient conditions for the rotational-add diffusion layers to attain the MDS property. These conditions enable explicit characterization of suitable primes p for given parameters. Leveraging these results, we construct three distinct families of rotational-add MDS diffusion layers applicable to AO ciphers. Although a rotational-add diffusion layer with 7 rotations and only additions has already been used in the design of the FHE-friendly block cipher YuX recently, to our knowledge, our work presents the first systematic theoretical characterization of rotational-add MDS diffusion layers and provides explicit constructions of them.

Keywords: Arithmetization-oriented symmetric cipher · Rotational-add diffusion layer · MDS matrix · Circulant matrix · Determinant

1 Introduction

Originally proposed by Shannon, the principles of confusion and diffusion remain fundamental to both the theory and implementation of symmetric ciphers. The substitution-permutation network (SPN) framework—a widely adopted structure in the design of symmetric ciphers—achieves confusion through nonlinear S-boxes and diffusion via linear layers (commonly termed linear diffusion layers). In designing these layers for symmetric ciphers, the primary goal is to enhance resistance against the two most significant cryptanalytic techniques: differential attacks and linear attacks [DR02]. This paper focuses on the design of effective linear diffusion layers that also feature low computational overhead and high implementation efficiency. However, our target application is primarily arithmetization-oriented (AO) symmetric ciphers instead of conventional ones. AO symmetric ciphers represent a relatively new research area in symmetric cryptography. A main difference between AO ciphers and conventional ones is that they are typically defined over a general finite field $\mathbb{F}_p$ instead of $\mathbb{F}_2$.

From a security perspective, the branch number of a linear diffusion layer plays a central role in determining the cipher's resistance to differential and linear attacks [DR02]. Higher branch numbers signify stronger diffusion, thereby substantially increasing the complexity of such attacks. When both the differential and linear branch numbers reach their theoretical maximum, the diffusion layer is termed a maximum distance separable (MDS) diffusion layer, corresponding to an MDS matrix in its matrix representation. MDS matrices provide optimal diffusion properties, enabling symmetric ciphers to achieve robust resistance against differential and linear attacks.

Beyond security considerations, the implementation cost of the diffusion layer is also a critical factor. Diffusion layers typically involve scalar multiplications, which can lead to considerable computational overhead in AO symmetric cipher implementations. Particularly in fully homomorphic encryption (FHE) applications, scalar multiplications can cause remarkable noise growth, which remains a technical bottleneck of current FHE schemes. For instance, FHE-friendly ciphers like Masta [HKC+20], Pasta [DGH+23] and Pasta v2 [GLR+24] experience notable noise accumulation due to extensive scalar multiplications in their diffusion layers. However, most of existing designs have not focused on optimizing this aspect to mitigate noise growth [HL20, HKC+20, CIR22, DGH+23, GLR+24]. Therefore, it is crucial to develop linear diffusion layers that minimize or eliminate scalar multiplications. Such designs not only improve implementation efficiency but also enhance the practicality of AO ciphers.

Despite their significance, systematic studies on constructing MDS diffusion layers for AO ciphers remain scarce in the literature, whereas substantial work on this topic exists for conventional symmetric ciphers in recent years. Current MDS diffusion layer constructions can be broadly categorized into two classes: matrix-based single-cycle constructions and circuit-based iterative constructions. Matrix-based single-cycle constructions directly design an MDS matrix that can be implemented within a single clock cycle. Existing research in this area focuses on building diffusion layers using special matrix classes over finite fields such as Cauchy matrices [RL89], Hadamard matrices [SDMO12, SKOP15] and Toeplitz matrices [SS16]. The primary advantage of this approach lies in low latency of the diffusion layers. However, it entails relatively high storage overhead as a $t \times t$ MDS matrix requires storage of t^2 elements when ignoring potential matrix structure optimizations.

On the other hand, circuit-based iterative constructions build an MDS diffusion layer by repeatedly applying a low-complexity circuit with small gate count and shallow depth. A commonly adopted technique in this category is recursive construction, first introduced in the design of PHOTON hash function [GPP11] and later applied to block ciphers such as LED [GPPR11]. This approach essentially constructs an MDS matrix by exponentiating a sparse matrix (actually, a companion matrix). Such a diffusion layer can be efficiently implemented by clocking a linear feedback shift register (LFSR) over several cycles. Other iterative constructions consist of (generalized) Feistel-structures-based approaches [WWW13, GWG16] and coding-theory-based approaches [Ber13, AF15, GPV17]. The main advantage of iterative diffusion layers is their small storage requirement. However, this comes at the cost of increased latency, as sequential clock cycles are necessary to complete the linear transformation.

Among existing MDS diffusion layer constructions, circulant matrices play a pivotal role. A circulant matrix is determined by its first row, with each subsequent row generated via a circulant shift. This structure allows implementations to instantiate only a single row of the matrix. For matrix-vector multiplication, the input vector can be circulantly shifted each clock cycle and fed into the same circuit to produce the output, enabling substantial circuit reuse and reducing implementation costs. Thanks to these advantages, circulant-matrix-based diffusion layers offer an attractive trade-off between efficiency and latency.

In constructing diffusion layers for symmetric ciphers, two types of circulant matrices are considered: block-wise circulant matrices and bit-wise circulant matrices. A block-wise circulant matrix is defined over the matrix space $\mathbb{F}_2^{m\times m}$, namely, it is the so-called block-circulant matrix [Dav94]. In concrete constructions, circulant matrices over the extension field $\mathbb{F}_{2^m}$ are often considered, which corresponds to block-circulant matrices whose blocks are pairwise commutative via matrix representations of finite field elements. A representative example is the column mixing matrix of AES, which is a 4×4 circulant matrix over $\mathbb{F}_{2^8}$. Constructions based on general block-wise circulant matrices are investigated in [LW16]. In this work, blocks that are not necessarily pairwise commutative are considered to produce lightweight matrices over $\mathbb{F}_2^{4\times 4}$ and $\mathbb{F}_2^{8\times 8}$, yielding lighter MDS matrices unattainable via the extension-field approach.

A bit-wise circulant MDS matrix is defined over $\mathbb{F}_2$. In symmetric cipher designs, this is often instantiated as a diffusion layer using only rotations and XOR operations, usually termed a rotational-XOR diffusion layer in the literature. Several cryptographic standards, such as the block cipher SM4 [ISO21] and the stream cipher ZUC [ISO20], employ this kind of construction. In [GLG+17], the authors perform a theoretic study on classifications and constructions of rotational-XOR MDS diffusion layers over $(\mathbb{F}_2^m)^4$, proving that the minimal possible size of the rotation offset is 5 subject to the MDS property and specifying an explicit class of MDS diffusion layers under this constraint. However, we identify some flaws in Theorem 5 and Theorem 6 of this paper.

Despite different techniques and significant developments in constructing MDS diffusion layers for conventional symmetric ciphers, generalizing these to adjust the design of AO ciphers remains unexplored. Since most AO ciphers are designed over $\mathbb{F}_p$ for odd prime p, direct generalizations of conventional MDS matrices seem unattainable. Specially for rotational-XOR diffusion layers, a natural research problem arises, that is, how to define their analogies over $\mathbb{F}_p$ and study classifications and explicit constructions of MDS ones therein. Note that XOR operation is unavailable over $\mathbb{F}_p$ anymore; instead, additions/subtractions and scalar multiplications modulo p should be considered. Crucially, non-trivial scalar multiplications will introduce additional algebraic complexity compared to the binary case.

The only known attempt in this direction appears in the design of the FHE-friendly symmetric cipher YuX [LLC+24], which utilizes an MDS diffusion layer over $(\mathbb{F}_p^4)^4$ generalizing the conventional rotational-XOR diffusion layer by replacing the XOR operation with modulo addition. Notably, it avoids non-trivial scalar multiplications—an attractable feature in FHE application settings. The rotation offset has a size 7, which was obtained by exhaustive search. However, our experimental results suggest that this rotation offset is not optimal and the size can be further reduced to 5, indicating that the diffusion layer used in YuX may potentially be improved by an alternative that requires fewer operations. In addition, it is worth noting that we can also search out MDS diffusion layers incorporating modulo subtraction operations, which avoid non-trivial scalar multiplications as well.

Our Contribution. Motivated by the YuX diffusion layer and the work on rotational-XOR diffusion layers in [GLG+17], this paper introduces and studies the rotational-add diffusion layers over $\mathbb{F}_p$ for arbitrary prime p. They coincide with the conventional rotational-XOR diffusion layers when $p = 2$. Rotational-add diffusion layers utilize only rotations and modular additions/subtractions, thus can be efficiently implemented via simple circuits. Crucially, since the scalars involved are limited to ± 1, the total cost of scalar multiplications can be reduced. This makes such diffusion layers exceptionally well-suited for AO symmetric ciphers, especially for FHE-friendly ciphers.

Our contributions are threefold:

1. To our knowledge, this work provides the first systematic treatment of MDS diffusion layers over $\mathbb{F}_p$ (for arbitrary prime p) targeting applications in AO symmetric ciphers. The introduced rotational-add diffusion layers generalize the classical rotational-XOR diffusion layers, offering new candidates for the design of diffusion layers of AO ciphers. Hopefully, this work can inspire further research on constructing MDS diffusion layers for specific application scenarios.

2. We characterize rotational-add diffusion layers that are MDS. For a rotational-add diffusion layer over $(\mathbb{F}_p^m)^n$, it is proved that the size of the rotation offset is lower bounded by $n+1$ under the MDS condition. In the case $n=4$ and no -1 scalar appears, we establish some non-existence results on rotational-add MDS diffusion layers with rotation offsets of size 5. Although a full classification of such optimal rotational-add MDS diffusion layers has not yet been achieved, we specify a large class among them, from which the study can be divided into 16 cases by traversing all possible patterns of appearance of the scalars ± 1. It turns out that MDS diffusion layers exist in all these 16 cases.

3. Extracting out 4 special cases from these 16 cases, we derive necessary and sufficient conditions on MDS properties of the constructed rotational-add diffusion layers over $(\mathbb{F}_p^m)^4$ in term of resultants of polynomials. This yields a general technique for constructing explicit classes of rotational-add MDS diffusion layers. As a result, three example classes are exhibited, and their inverse diffusion layers are derived as well. Experimental results[1] demonstrate that when applied in YuX, the efficiency of our rotational-add MDS diffusion layers outperform the original one proposed in [LLC+24] in FHE application settings. As a byproduct, we rectify the flaws in [GLG+17, Theorems 5 and Theorem 6].

Organization. The rest of the paper is organized as follows. Section 2 presents some basic notations and background knowledge that will be used throughout the paper, together with the formal definition of rotational-add diffusion layers. Section 3 presents several necessary conditions for rotational-add MDS diffusion layers and properties of two important classes of matrices which form the main toolkit for our subsequent discussions. In Section 4, we try to classify rotational-add MDS diffusion layers over $(\mathbb{F}_p^m)^4$ with 5 rotations, reducing our study to one special class. In Section 5, we establish necessary and sufficient conditions for four types of rotational-add diffusion layers to achieve the MDS property. Three classes of explicit constructions and their inverses are also obtained, followed by a brief discussion on their efficiency. Conclusions and potential future research directions are given in Section 6.

2 Preliminaries

In this section, we firstly fix some basic notations that will be used throught the paper. Then, we recall some notions and results from linear algebra. Next, we give an introduction to the concepts of branch numbers of linear diffusion layers in symmetric ciphers and MDS matrices, followed by their properties. At last, we propose a special type of linear diffusion layer, the rotational-add diffusion layer over arbitrary finite prime field, which is the main target of our current research.

[1] Codes available at https://github.com/ba0fengwu/rotadd_mds_diffusion_layers

2.1 Notations

Unless otherwise specified, all indices are assumed to start from zero throughout the following discussions.

- $|S|$: cardinality of a set S;
- $\mathbb{F}_q$: finite field of order q, where q is a prime or a prime power;
- $\boldsymbol{x} \lll k$: left rotation of a vector $\boldsymbol{x}$ by k positions;
- $\mathrm{wt}(\boldsymbol{x})$: Hamming weight of a vector $\boldsymbol{x}$, i.e., the number of nonzero entries of $\boldsymbol{x}$;
- $\boldsymbol{e}_i$: standard basis vector of a vector space, i.e., a vector with only one nonzero entry 1 at the i-th position;
- $R^{n\times n}$: the ring of $n \times n$ matrices over a ring R;
- $GL(m, R)$: the general linear group of order m over a commutative ring R, i.e., the group of all $m \times m$ nonsingular matrices over R;
- $\det M$ or $\det(M)$: determinant of a matrix M;
- M^τ: tranpose of a matrix M;
- $\mathrm{Diag}(a_0, a_1, \ldots, a_{n-1})$: diagonal matrix with diagonal entries $a_0, a_1, \ldots, a_{n-1}$;
- I_k: identity matrix of order k.

2.2 Linear-algebraic notions and results

We recall some notions and results from linear algebra, which can be found in many textbooks such as [BF73, HJ91].

Let n be a positive integer. An upper shift matrix of order n is a matrix of the form

$$U_n = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ 0 & 0 & 0 & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix},$$

that is, U_n is a binary matrix with ones on the first superdiagonal and zeros elsewhere. The transpose of U_n is called a lower shift matrix, denoted by L_n. When it is clear from the context, we shall omit the subscript n. The following properties of the upper and lower shift matrices are easy to derive.

Proposition 1. *1. For any $1 \le k \le n-1$, the matrix U^k has entries 1 on the k-th superdiagonal and 0 elsewhere, and L^k is the transpose of U^k. Therefore, both U and L are nilpotent matrices, more precisely, we have $U^n = L^n = 0$;*

2. $ULU = U$, $LUL = L$;

3. $UL = I_n - \mathrm{Diag}(0, 0, \ldots, 1)$, $LU = I_n - \mathrm{Diag}(1, 0, \ldots, 0)$. So, UL and LU are both symmetric.

In fact, the second and third parts of Proposition 1 indicates that U and L are Moore-Penrose pseudo-inverses of each other.

The matrix $P = U + L^{n-1}$ is called a cyclic permutation matrix of order n. Actually, it is the matrix representation of the linear transformation $\lll 1$ over an n-dimensional vector space. For any $1 \le k \le n-1$, P^k is also called a cyclic permutation matrix, representing the $\lll k$ operation. Therefore, we have $P^k = U^k + L^{n-k}$ and $P^n = I$, so the inverse of P is P^{n-1}, which equals P^τ. It follows that:

Proposition 2. *For any $1 \le k \le n-1$, let $d = \gcd(k, n)$. Then the characteristic polynomial and minimal polynomial of the cyclic permutation matrix P^k is $(x^{n/d} - 1)^d$ and $x^{n/d} - 1$, respectively. In particular, if $d = 1$, then the minimal polynomial of P^k is $x^n - 1$, which coincides with its characteristic polynomial.*

For two matrices A and B, their Kronecker product or tensor product is defined as

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix} \text{ where } A = (a_{ij})_{1\le i\le m,\ 1\le j\le n}.$$

For the cyclic permutation matrix P_n^k, $1 \le k \le n-1$, we can find that $P_n^k = P_{n/d}^{k/d} \otimes I_d$ where $d = \gcd(k, n)$. This can be easily observed by partitioning P_n^k into a block matrix with $d \times d$ blocks. Note that $\gcd(k/d, n/d) = 1$. It means that when dealing with $n \times n$ cyclic permutation matrices, sometimes we can use the tensor decomposition to reduce our target to the primitive case, that is, cyclic permutation matrices with smaller sizes having same characteristic and minimal polynomials. In fact, to prove Proposition 2, we need this trick and the fact that

$$\det(A \otimes B) = \det(A)^v \cdot \det(B)^u \tag{1}$$

if A and B are $u \times u$ and $v \times v$ square matrices, respectively.

The following elementary results on determinants of 2×2 block matrices will be frequently used in our study.

Proposition 3. *Let A, B, C and D be $m \times m$, $m \times n$, $n \times m$ and $n \times n$ matrices over a field $\mathbb{F}$, respectively. Then, the following properties hold:*

1. *if A is nonsingular, then*

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(A) \cdot \det(D - CA^{-1}B).$$

The matrix $D - CA^{-1}B$ is called the Schur complement of the block A;

2. *If $m = n$ and $CD = DC$, then*

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - BC);$$

3. *If $m = n$, then*

$$\det \begin{pmatrix} A & B \\ B & A \end{pmatrix} = \det(A + B) \cdot \det(A - B).$$

As for determinants of general block matrices, the following result is very useful.

Theorem 1 ([Bou89]). *Let R be a commutative ring and let S be a commutative subring of $R^{m\times m}$. Then for any matrix $M \in S^{n\times n} \subseteq R^{mn\times mn}$, we have*

$$\det_R(\det_S(M)) = \det_R(M).$$

Here the subscript of det *is used to indicate the ring over which the determinant is computed.*

Theorem 1 says that the determinant of a block matrix over a commutative ring R whose blocks are pairwise commutative, can be computed by first computing the determinant viewing each block as a single element to obtain a matrix over R, and then taking the determinant of this resulting matrix. This is particularly useful when dealing with matrices that can be partitioned into block matrices with pairwise-commutative blocks, allowing us to simplify the computation of their determinants.

For two polynomials $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{m} b_i x^i$ over a field $\mathbb{F}$, their resultant is defined as determinant of the Sylvester matrix, that is,

$$\mathrm{Res}(f,g) = \det \begin{pmatrix} a_n & a_{n-1} & \cdots & a_0 & & & & \\ & a_n & a_{n-1} & \cdots & a_0 & & & \\ & & \ddots & \ddots & & & \ddots & \\ & & & a_n & a_{n-1} & & \cdots & a_0 \\ b_m & b_{m-1} & \cdots & b_0 & & & & \\ & b_m & b_{m-1} & \cdots & b_0 & & & \\ & & \ddots & \ddots & & & \ddots & \\ & & & b_m & b_{m-1} & & \cdots & b_0 \end{pmatrix} \begin{array}{l} \left.\vphantom{\begin{matrix} a\\a\\a\\a \end{matrix}}\right\} m \text{ rows} \\ \left.\vphantom{\begin{matrix} a\\a\\a\\a \end{matrix}}\right\} n \text{ rows} \end{array} .$$

The following result is useful in determining whether f and g have common roots.

Proposition 4. *Let $f(x)$ and $g(x)$ be two polynomials over a field $\mathbb{F}$. Then $\gcd(f,g) = 1$ if and only if $\mathrm{Res}(f,g) \neq 0$.*

At last, we recall some knowledge about circulant and block-circulant matrices. For details, we refer to [Dav94].

Definition 1. Let $(a_0, a_1, ..., a_{n-1})$ be a vector over a ring R. An $n \times n$ matrix, denoted by $\mathrm{Circ}(a_0, a_1, ..., a_{n-1})$, is called a *circulant matrix* if the (i,j)-th entry is given by a_{j-i}, where $0 \leq i, j \leq n-1$ and the subscripts are taken modulo n. In particular, if $R = S^{m\times m}$ for a ring S, then $\mathrm{Circ}(a_0, a_1, ..., a_{n-1})$ can be viewed as an $n \times n$ block matrix with $m \times m$ blocks over S, and is often known as a block-circulant matrix.

We are mainly interested in circulant or block-circulant matrices over a field $\mathbb{F}$. In this case, they have many nice properties. For example, for $(a_0, a_1, ..., a_{n-1}) \in \mathbb{F}^n$, we know that $\mathrm{Circ}(a_0, a_1, ..., a_{n-1})$ is nonsingular if and only if

$$\gcd\left(\sum_{i=0}^{n-1} a_i x^i,\ x^n - 1\right) = 1.$$

In fact, all $n \times n$ circulant matrices over $\mathbb{F}$ form a commutative ring, which is isomorphic to the polynomial residue ring $\mathbb{F}[x]/(x^n - 1)$. Under this isomorphism, every circulant matrix has an associated polynomial, whose coefficients are the entries of the first row of the matrix. Therefore, if $\mathrm{Circ}(a_0, a_1, ..., a_{n-1})$ is nonsingular, then its inverse can be obtained from coefficients of the inverse polynomial of $\sum_{i=0}^{n-1} a_i x^i$ modulo $x^n - 1$.

For singularity of block-circulant matrices over $\mathbb{F}$, we only mention a result in the case $n = 4$, that is, for

$$\mathrm{Circ}(A, B, C, D) = \begin{pmatrix} A & B & C & D \\ D & A & B & C \\ C & D & A & B \\ B & C & D & A \end{pmatrix}, \tag{2}$$

where $A,\ B,\ C,\ D \in \mathbb{F}^{m\times m}$, we have

$$\begin{aligned} \det \mathrm{Circ}(A, B, C, D) = &\det(A + B + C + D) \\ &\cdot \det(A - B + C - D) \\ &\cdot \det\left(A + \sqrt{-1}B - C - \sqrt{-1}D\right) \\ &\cdot \det\left(A - \sqrt{-1}B - C + \sqrt{-1}D\right). \end{aligned} \tag{3}$$

Here $\sqrt{-1}$ is a square root of -1 which lies in $\mathbb{F}$ or a quadratic extension of $\mathbb{F}$. Note here that this formula is valid for any $A,\ B,\ C,\ D$ over $\mathbb{F}$, not necessarily pairwise commutative.

2.3 Branch numbers of linear diffusion layers

In an SPN-based symmetric cipher, the substitution layer typically consists of n parallel S-boxes, each acting on a word of length m over a finite field $\mathbb{F}_p$, to provide nonlinear properties for the cipher. The diffusion layer of the cipher aims to mix the outputs of these S-boxes and is generally realized as an invertible linear/affine transformation over $(\mathbb{F}_p^m)^n$. The branch number of the diffusion layer can evaluate the cipher's resistance to the two most important cryptanalytic techniques, namely, the differential and linear attacks [DR02]. A higher branch number implies greater diffusion and consequently increases the difficulty of performing such attacks. In the following, we give the definition of differential branch number and linear branch number of a linear diffusion layer.

We use $\boldsymbol{x} = (\boldsymbol{x}_0, \boldsymbol{x}_1, \ldots, \boldsymbol{x}_{n-1}) \in (\mathbb{F}_p^m)^n$ to represent a compound vector, where $\boldsymbol{x}_i \in \mathbb{F}_p^m$, $0 \le i \le n-1$, is called a bundle or a word. The corresponding output of a diffusion layer $L_{n,m}$ is expressed as $L_{n,m}(\boldsymbol{x})$. It is simply $L_{n,m} \cdot \boldsymbol{x}^\tau$ if $L_{n,m}$ is in its matrix representation (viewed as an $n \times n$ block matrix with $m \times m$ blocks).

Definition 2. Let $L_{n,m}$ be a linear diffusion layer over $(\mathbb{F}_p^m)^n$.

1. The differential branch number of $L_{n,m}$ is defined as

$$\mathcal{B}_d(L_{n,m}) = \min_{\boldsymbol{x} \in (\mathbb{F}_p^m)^n, \boldsymbol{x} \neq \boldsymbol{0}} \{\mathrm{wt}(\boldsymbol{x}) + \mathrm{wt}(L_{n,m}(\boldsymbol{x}))\}.$$

2. The linear branch number of $L_{n,m}$ is defined as

$$\mathcal{B}_\ell(L_{n,m}) = \min_{\boldsymbol{x} \in (\mathbb{F}_p^m)^n, \boldsymbol{x} \neq \boldsymbol{0}} \{\mathrm{wt}(\boldsymbol{x}) + \mathrm{wt}(L_{n,m}^\tau(\boldsymbol{x}))\}.$$

Note here that $\mathrm{wt}(\boldsymbol{x})$ represents the number of nonzero bundles in $\boldsymbol{x}$, so sometimes it is called the bundle weight of $\boldsymbol{x}$.

It is easy to see that both $\mathcal{B}_d(L_{n,m})$ and $\mathcal{B}_\ell(L_{n,m})$ are upper bounded by $n+1$. $L_{n,m}$ is called a perfect or *maximum distance separable (MDS)* diffusion layer if this upper bound is attained [Vau95], that is, $\mathcal{B}_d(L_{n,m}) = \mathcal{B}_\ell(L_{n,m}) = n + 1$. The corresponding matrix is called an MDS matrix.

Generally, it is inconvenient to determine whether a linear diffusion layer is MDS via computing the differential and linear branch numbers. However, the following pure linear-algebraic characterization is available.

Theorem 2 ([BR99])**.** *Let $L_{n,m}$ be an $n \times n$ block matrix with $m \times m$ blocks. Then $L_{n,m}$ is MDS if and only if every k-order block submatrix is nonsingular for all $1 \leq k \leq n$.*

From Theorem 2 we can easily get the following results, which have also been frequently used in some previous works; see [LS16, GLG+17, TTKS18].

Proposition 5. *Let $L_{n,m}$ be an $n \times n$ block matrix with $m \times m$ blocks. Then $L_{n,m}$ is MDS if and only if $L_{n,m}^{\tau}$ is MDS.*

Proposition 5 indicates that a diffusion layer attains the maximum differential branch number if and only if it also achieves the maximum linear branch number. Therefore, in the following when talking about MDS diffusion layers, we omit the distinction between the two and refer to both collectively as the branch number.

Proposition 6. *Let $L_{n,m}$ be an $n\times n$ block matrix with $m\times m$ blocks, and let P and Q be two permutation matrices of order n. Then $L_{n,m}$ is MDS if and only if $(P \otimes I_m) \cdot L_{n,m} \cdot (Q \otimes I_m)$ is MDS.*

Proposition 7. *Let $L_{n,m}$ be an $n \times n$ block matrix with $m \times m$ blocks, and let D_1 and D_2 be two nonsingular block diagonal matrices of order n with $m \times m$ diagonal blocks. Then $L_{n,m}$ is MDS if and only if $D_1 \cdot L_{n,m} \cdot D_2$ is MDS.*

Proposition 6 and Proposition 7 indicate that the MDS property of a linear diffusion layer is invariant under the action of permutating the block rows and columns of the diffusion matrix, and the action of multiplying each block row/column by a same invertible block. These kinds of equivalences between MDS matrices can help us to construct new MDS matrices from old ones or reduce the search space when searching for MDS matrices.

Although MDS matrices are useful in the design of symmetric ciphers with strong diffusion properties, in practice it is not so easy to construct and search for them, especially for those with favorable implementation characteristics. In the next subsection, we will introduce a special class of linear diffusion layers with good implementation traits in the design of AO symmetric ciphers, and this paper is devoted to studying MDS properties of such diffusion layers.

2.4 The rotational-add diffusion layer

We consider diffusion layers over the vector space $(\mathbb{F}_p^m)^n$ constructed by only left-rotation and addition/subtraction operations for arbitrary prime p.

Definition 3. Let n, m be positive integers and $\mathcal{G}$ be a proper subset of $\{0, 1, \ldots, nm-1\}$. Assume $\mathcal{G} = \mathcal{I} \cup \mathcal{J}$ with $\mathcal{I} \cap \mathcal{J} = \emptyset$. A rotational-add diffusion layer over $(\mathbb{F}_p^m)^n$ determined by $\mathcal{G}$, denoted by $L_{n,m}^{\mathcal{G}}$, is defined as

$$L_{n,m}^{\mathcal{G}}(\boldsymbol{x}) = \sum_{i \in \mathcal{I}} (x_0, x_1, \ldots, x_{nm-1}) \lll i - \sum_{j \in \mathcal{J}} (x_0, x_1, \ldots, x_{nm-1}) \lll j, \tag{4}$$

where $\boldsymbol{x} = (x_0, x_1, \ldots, x_{nm-1}) \in (\mathbb{F}_p^m)^n$.

Note that in the definition of $L_{n,m}^{\mathcal{G}}$, the addition and subtraction operations are performed over the finite field $\mathbb{F}_p$, so when $p = 2$, they coincide with the bitwise XOR operation. In this case, the rotational-add diffusion layer is also known as a rotational-XOR diffusion layer in some previous works [GLG+17]. This class of diffusion layers can offer a favorable trade-off between area requirement and clock cycle latency, making it particularly well-suited for balanced hardware implementations. Therefore, they have been widely used in the designs of some important symmetric ciphers, such as the block cipher SM4 [ISO21], the stream cipher ZUC [ISO20], the hash function SHA-256 [Nat15], the authenticated encryption algorithm Ascon [DEMS21], etc.

In [GLG+17], the authors systematically investigate the conditions under which the rotational-XOR diffusion layers can be MDS in the case $n = 4$. They also construct a wide class of rotational-XOR MDS diffusion layers by specifying the ratotion offset $\mathcal{G}$ with a size 5, which is the theoretical minimum. However, when p is an odd prime, to the best of our knowledge, no systematic research on characterizations and constructions of rotational-add MDS diffusion layers has been conducted before. In [LLC+24], a rotational-add MDS diffusion layer is utilized in the design of the FHE-friendly block cipher YuX for the first time, where $n = 4$ and a rotation offset with size 7 is applied. The advantage of using a rotational-add diffusion layer in an AO symmetric cipher is that, the cost of scalar multiplications can be reduced since its matrix representation contains only 0 and ± 1 in $\mathbb{F}_p$. A natural remaining question is whether it is possible to construct rotational-add MDS diffusion layers with smaller rotation offsets, and in praticular, with the minimum achievable size.

To answer this question, it is reasonable to choose the case $n = 4$ as a starting point, since when designing symmetric ciphers utilizing rotational-add diffusion layers, they are usually applied on 4 intermediate state words. In fact, the YuX cipher falls into this category. From the definition of the rotational-add diffusion layer $L_{4,m}^{\mathcal{G}}$, it is obvious that its matrix representation is a $4m \times 4m$ circulant matrix over $\mathbb{F}_p$ with only $(0, 1, -1)$-entries, in which the places of nonzero entries of the first row is determined by the rotation offset $\mathcal{G}$. When talking about MDS property of $L_{4,m}^{\mathcal{G}}$, we should view it as a 4×4 block matrix with $m \times m$ blocks, which is naturally a block circulant matrix. More precisely,

$$L_{4,m}^{\mathcal{G}} = \mathrm{Circ}(A, B, C, D) = \begin{pmatrix} A & B & C & D \\ D & A & B & C \\ C & D & A & B \\ B & C & D & A \end{pmatrix}, \tag{5}$$

where the blocks A, B, C and D are all $m \times m$ matrices containing only 0 and ± 1. Since $L_{4,m}^{\mathcal{G}}$ is itself a circulant matrix over $\mathbb{F}_p$, each of these four blocks must have this form: all nonzero elements appear only on the main diagonal or certain superdiagonals or subdiagonals, whose indices are determined by $\mathcal{G}$, and entries of each nonzero diagonal are the same (1 or -1).

For example, when $m = 4$, $\mathcal{G} = \{0, 2, 6, 10, 12\}$ with $\mathcal{I} = \{6, 10, 12\}$ and $\mathcal{J} = \{0, 2\}$, the corresponding $L_{4,4}^{\mathcal{G}}$ looks like

$$\left(\begin{array}{cccc:cccc:cccc:cccc}
-1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\
0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\
\hdashline
1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\
0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\
1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\
\hdashline
0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 \\
1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 \\
\hdashline
0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & -1 \\
-1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\
0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1
\end{array}\right).$$

3 Basic properties of rotational-add diffusion layers

In this section, we investigate some properties of the rotational-add diffusion layers and establish some necessary conditions for them to be MDS. These results can serve as a foundation of the subsequent constructions of such MDS diffusion layers and designs of new symmetric ciphers.

Let $L^{\mathcal{G}}_{n,m}$ be a rotational-add diffusion layer defined in (4). Abusing notations, we also use $L^{\mathcal{G}}_{n,m}$ to denote its matrix representation, which is an $n \times n$ block matrix with $m \times m$ blocks. By Proposition 7, we can assume $|\mathcal{I}| \geq |\mathcal{J}|$ without loss of generality when talking about the MDS property of $L^{\mathcal{G}}_{n,m}$. Note that an extreme case is $|\mathcal{J}| = 0$, which will be discussed in detail in the next section.

Proposition 8. *If $L^{\mathcal{G}}_{n,m}$ is nonsingular, then we have $p \nmid (|\mathcal{I}| - |\mathcal{J}|)$.*

Proof. Acting $L^{\mathcal{G}}_{n,m}$ on $\boldsymbol{x} = (1, 1, \ldots, 1)$, we have

$$L^{\mathcal{G}}_{n,m}(\boldsymbol{x}) = (|\mathcal{I}| - |\mathcal{J}|) \cdot \boldsymbol{x}.$$

Therefore, $p \mid (|\mathcal{I}| - |\mathcal{J}|)$ is equivalent to $L^{\mathcal{G}}_{n,m}(\boldsymbol{x}) = \mathbf{0}$, which implies that $L^{\mathcal{G}}_{n,m}$ is singular sicne $\boldsymbol{x}$ is a nonzero vector. □

Remark 1. Proposition 8 says that $p \nmid (|\mathcal{I}| - |\mathcal{J}|)$ is a necessary condition for $L^{\mathcal{G}}_{n,m}$ to be MDS according to Theorem 2. In particular, when $p = 2$, i.e., when $L^{\mathcal{G}}_{n,m}$ is defined over $(\mathbb{F}_2^m)^n$, $|\mathcal{G}|$ must be odd for $L^{\mathcal{G}}_{n,m}$ to be MDS.

By Theorem 2, we can also derive a lower bound of $|\mathcal{G}|$ for $L^{\mathcal{G}}_{n,m}$ to be MDS at first.

Proposition 9. *Assume $n > 1$. If $L^{\mathcal{G}}_{n,m}$ is an MDS matrix, then we have $|\mathcal{G}| \geq n + 1$.*

Proof. Assume $L^{\mathcal{G}}_{n,m} = \mathrm{Circ}(M_0, M_1, \ldots, M_{n-1})$, where M_i, $0 \leq i \leq n-1$, are the first block-row of $L^{\mathcal{G}}_{n,m}$. Recall that for any $0 \leq k,\ j \leq n-1$, the (k, j)-th block of $L^{\mathcal{G}}_{n,m}$ is M_{j-k} (all subscripts are taken modulo n here and below). From Definition 3, we see that the rotation offset $\mathcal{G}$ determines the first row of M_i for each $0 \leq i \leq n-1$.

We proceed by proving if $|\mathcal{G}| \leq n$, there exists a block submatrix of $L^{\mathcal{G}}_{n,m}$ that is singular, then according to Theorem 2, $L^{\mathcal{G}}_{n,m}$ is not MDS.

When $|\mathcal{G}| < n$, there must exist one i such that the first row of M_i is 0, and then M_i is a zero matrix or a lower triangular matrix with a zero main diagonal, which must be singular.

When $|\mathcal{G}| = n$ and there exists one i such that the first row of M_i is 0, then M_i is singular as discussed above. Otherwise, assume $\mathcal{G} = \{g_0,\ g_1,\ \ldots,\ g_{n-1}\}$ satisfying $im \leq g_i < (i+1)m$ for $0 \leq i \leq n-1$. If there exists $0 \leq i \leq n-2$ s.t. $g_{i+1} - g_i > m$, WLOG, assuming $g_1 - g_0 > m$, one can figure out that M_1 must contain a zero row, thus it is singular. So, we have $g_{i+1} - g_i \leq m$ for $0 \leq i \leq n-2$, but if $g_{i+1} - g_i < m$ for one certain i, it follows then $nm - g_{n-1} > m - g_0$, which yields M_0 is singular. Therefore, the only possibility is $\mathcal{G} = \{g_0,\ g_0 + m,\ g_0 + 2m,\ \ldots,\ g_0 + (n-1)m\}$ where $g_0 \geq 0$. Under this condition, we can see that all the M_i, $0 \leq i \leq n-1$, have the same shape. More precisely, assume the first row of M_i has a unique nonzero entry $a_i \in \{\pm 1\}$. Then we have

$$M_i = \mathrm{Diag}(a_i, \ldots, a_i, a_{i-1}, \ldots, a_{i-1}) \cdot P_m^{g_0} := D_i \cdot P_m^{g_0},$$

where a_i appears $m - g_0$ times in the diagonal matrix D_i and P_m is the cyclic permutation matrix of order m, $0 \leq i \leq n-1$. Now the block matrix $L^{\mathcal{G}}_{n,m}$ can be represented as

$$L^{\mathcal{G}}_{n,m} = \mathrm{Circ}(D_0, D_1, \ldots, D_{n-1}) \cdot \mathrm{Diag}(P_m^{g_0}, \ldots, P_m^{g_0}).$$

We need only to extract out singular submatrices of $D_{n,m} = \mathrm{Circ}(D_0, D_1, \ldots, D_{n-1})$ to finish the proof according to Proposition 7.

If n is even, for any $0 \leq i < n/2$, $D_{n,m}$ has a 2×2 submatrix of the form

$$\begin{pmatrix} D_i & D_{i+n/2} \\ D_{i-n/2} & D_i \end{pmatrix} = \begin{pmatrix} D_i & D_{i+n/2} \\ D_{i+n/2} & D_i \end{pmatrix},$$

whose determinant is $\det(D_i^2 - D_{i+n/2}^2) = \det(I_m - I_m) = 0$ according to Theorem 1.

If n is odd, we assume $a_j = a_k$ for certain $0 \leq j < k \leq n-1$. Let $i = (j+k)/2 \mod n$. Then the $(j-i, j)$- and $(j-i, i)$-th entries of $D_{n,m}$ are D_i and D_k, respectively. This means $D_{n,m}$ has a 2×2 submatrix of the form

$$\begin{pmatrix} D_i & D_j \\ D_k & D_i \end{pmatrix} \text{ or } \begin{pmatrix} D_j & D_i \\ D_i & D_k \end{pmatrix},$$

whose determinant is 0 since $D_i^2 = I_m$ and $D_j \cdot D_k$ has $m - g_0$ 1's on the diagonal. □

In the following analysis, we focus on the case $n = 4$ which is a common parameter size in symmetric cipher design as discussed before, and we investigate the MDS property of $L_{4,m}^{\mathcal{G}}$. Proposition 9 shows that the minimum size of the rotation offset $\mathcal{G}$ for $L_{4,m}^{\mathcal{G}}$ to be MDS is possibly 5. In the remaining of this paper, we focus on this optimal case and devote to characterizing and especially constructing some explicit classes of such MDS matrices.

Assume $|\mathcal{G}| = 5$ and $\mathcal{G} = \{g_1,\ g_2,\ g_3,\ g_4,\ g_5\}$. According to the proof of Proposition 9, in the representation (5) of $L_{4,m}^{\mathcal{G}}$, the blocks $A,\ B,\ C$ and D should have at least one nonzero element in each of their first rows for $L_{4,m}^{\mathcal{G}}$ to be MDS. WLOG, we can assume that the first row of A has two nonzero elements, that is to say,

$$0 \leq g_1 < g_2 < m \leq g_3 < 2m \leq g_4 < 3m \leq g_5 < 4m. \tag{6}$$

Otherwise, we can circulantly permute the block-rows of $L_{4,m}^{\mathcal{G}}$ to transform it into this form, which will keep the MDS property by Proposition 6. To explore the MDS property of $L_{4,m}^{\mathcal{G}}$, it needs only to check singularity of all of its block submatrics according to Theorem 2. To ease the subsequent discussions, in Proposition 10 we explicitly list all the block submatrics that should be considered, taking into account of the cyclic structure inherent in $L_{4,m}^{\mathcal{G}}$.

Proposition 10. *Let $L_{4,m}^{\mathcal{G}} = \mathrm{Circ}(A, B, C, D)$. Then the necessary and sufficient condition for it to be MDS is that the following submatrices are all nonsingular:*

1. *1-order submatrices: $A,\ B,\ C,\ D$;*
2. *2-order submatrices:*

$$\begin{pmatrix} A & B \\ D & A \end{pmatrix}, \begin{pmatrix} A & C \\ D & B \end{pmatrix}, \begin{pmatrix} A & D \\ D & C \end{pmatrix}, \begin{pmatrix} B & C \\ A & B \end{pmatrix}, \begin{pmatrix} B & D \\ A & C \end{pmatrix},$$

$$\begin{pmatrix} C & D \\ B & C \end{pmatrix}, \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \begin{pmatrix} A & C \\ C & A \end{pmatrix}, \begin{pmatrix} A & D \\ C & B \end{pmatrix}, \begin{pmatrix} B & D \\ D & B \end{pmatrix};$$

3. *3-order submatrices:*

$$\begin{pmatrix} A & B & C \\ D & A & B \\ C & D & A \end{pmatrix}, \begin{pmatrix} A & B & D \\ D & A & C \\ C & D & B \end{pmatrix}, \begin{pmatrix} A & C & D \\ D & B & C \\ C & A & B \end{pmatrix}, \begin{pmatrix} B & C & D \\ A & B & C \\ D & A & B \end{pmatrix};$$

4. *4-order submatrix, i.e., the whole matrix.*

Note that $L^{\mathcal{G}}_{4,m}$ has two 2×2 block submatrices of the special forms

$$\begin{pmatrix} A & C \\ C & A \end{pmatrix} \text{ and } \begin{pmatrix} B & D \\ D & B \end{pmatrix}.$$

As a result, according to Proposition 10, Proposition 3 and the formula (3), we can easily obtain some further necessary conditions for $L^{\mathcal{G}}_{4,m}$ to be MDS.

Corollary 1. *Let* $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, C, D)$*. If* $L^{\mathcal{G}}_{4,m}$ *is MDS, then the following matrices must be nonsingular:*

- $A,\ B,\ C,\ D$*;*
- $A + C,\ A - C,\ B + D,\ B - D$*;*
- $A + B + C + D,\ A - B + C - D$*;*
- $A + \sqrt{-1}B - C - \sqrt{-1}D,\ A - \sqrt{-1}B - C + \sqrt{-1}D$*.*

We investigate the 1-order submatrices firstly, that is, conditions for $A,\ B,\ C$ and D to be nonsingular. Note that $A,\ B,\ C$ and D are 2-diagonal or 3-diagonal matrices when $|\mathcal{G}| = 5$, which seem easy to deal with.

Definition 4 (2-diagonal matrix)**.** Let M be an $m \times m$ matrix over $\mathbb{F}_p$. It is called a 2-diagonal matrix if it has the form

$$M = c_1 \cdot U_m^{\alpha} + c_2 \cdot L_m^{\beta}$$

for some $0 \leq \alpha < m,\ 0 < \beta < m$ and $c_1,\ c_2 \in \mathbb{F}_p^*$, where U_m and L_m are the upper and lower shift matrices, respectively.

For example, when $m = 5,\ c_1 = 1,\ c_2 = -1,\ \alpha = 1$ and $\beta = 2$, the 2-diagonal matrix M looks like

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 0 \end{pmatrix}.$$

Definition 5 (3-diagonal matrix)**.** Let M be an $m \times m$ matrix over $\mathbb{F}_p$. It is called a 3-diagonal matrix if it has the form

$$M = c_1 \cdot U_m^{\alpha_1} + c_2 \cdot U_m^{\alpha_2} + c_3 \cdot L_m^{\beta}$$

for some $0 \leq \alpha_1 < \alpha_2 < m,\ 0 < \beta < m$ and $c_1,\ c_2,\ c_3 \in \mathbb{F}_p^*$, where U_m and L_m are the upper and lower shift matrices, respectively.

For example, when $m = 5,\ c_1 = c_2 = 1,\ c_3 = -1,\ \alpha_1 = 1,\ \alpha_2 = 2$ and $\beta = 3$, the 3-diagonal matrix M looks like

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ -1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 0 & 0 \end{pmatrix}.$$

In the following discussions, we adopt a unified notation for the diagonals. The α-th superdiagonal will be referred to as diagonal α, while the β-th subdiagonal will be referred to as diagonal $-\beta$. This is reasonable, because sometimes we write $U_m^{-\beta}$ instead of L_m^{β} as U_m is the Moore-Penrose pseudo-inverse of L_m.

Theorem 3. *Let M be an $m \times m$ 2-diagonal matrix over $\mathbb{F}_p$ with diagonals α, $-\beta$ and corresponding constants $c_1,\ c_2$. Then M is nonsingular if and only if $\alpha+\beta \mid m$ or $\alpha = 0$.*

Proof. If $\alpha = 0$, M is lower triangular and is obviously nonsingular. Hence, we focus our discussion on the case $\alpha > 0$. WLOG, we assume $\alpha \geq \beta$ since otherwise, we can consider M^τ. When $\alpha+\beta > m$, it is easy to see M is singular since it must contain a zero row. If $\alpha+\beta = m$, then we have

$$M = \begin{pmatrix} c_1 \cdot I_\beta & \\ & c_2 \cdot I_\alpha \end{pmatrix} \cdot P_m^\alpha,$$

where P_m^α is a cyclic permutation matrix. So, $\det(M) = c_1^\beta c_2^\alpha \cdot \det(P_m^\alpha) \neq 0$, which implies that M is nonsingular.

When $\alpha+\beta < m$, assume $m = k(\alpha+\beta)+t$ with $0 \leq t < \alpha+\beta$. We prove that M is nonsingular if and only if $t = 0$ through a recursive argument in the following. Firstly, we partition M along the $(\alpha+\beta-1)$-th column and the $(\alpha+\beta-1)$-th row into a block matrix of the form

$$M = \begin{pmatrix} N_{(\alpha+\beta)\times(\alpha+\beta)} & X_{(\alpha+\beta)\times(m-\alpha-\beta)} \\ Y_{(m-\alpha-\beta)\times(\alpha+\beta)} & \tilde{M}_{(m-\alpha-\beta)\times(m-\alpha-\beta)} \end{pmatrix}.$$

That is, we partition M by cutting out the top left corner N of it, which is an $(\alpha+\beta)\times(\alpha+\beta)$ submatrix. Then by Proposition 3, we have

$$\det(M) = \det(N) \cdot \det(\tilde{M} - YN^{-1}X).$$

From the above discussion, the block N is nonsingular and

$$N^{-1} = (P_{\alpha+\beta}^\alpha)^{-1} \cdot \begin{pmatrix} \frac{1}{c_1} \cdot I_\beta & \\ & \frac{1}{c_2} \cdot I_\alpha \end{pmatrix} = P_{\alpha+\beta}^\beta \cdot \begin{pmatrix} \frac{1}{c_1} \cdot I_\beta & \\ & \frac{1}{c_2} \cdot I_\alpha \end{pmatrix}.$$

Examing the blocks X and Y, we find they can further be partitioned into the forms

$$X = \begin{pmatrix} O_{\beta\times(m-\alpha-\beta)} \\ X'_{\alpha\times(m-\alpha-\beta)} \end{pmatrix} \quad \text{and} \quad Y = \begin{pmatrix} O_{(m-\alpha-\beta)\times\alpha} & Y'_{(m-\alpha-\beta)\times\beta} \end{pmatrix},$$

where O denotes the zero matrix. Then we have

$$\begin{aligned} YN^{-1}X &= \begin{pmatrix} O_{(m-\alpha-\beta)\times\alpha} & Y'_{(m-\alpha-\beta)\times\beta} \end{pmatrix} \cdot P_{\alpha+\beta}^\beta \cdot \begin{pmatrix} \frac{1}{c_1} \cdot I_\beta & \\ & \frac{1}{c_2} \cdot I_\alpha \end{pmatrix} \cdot \begin{pmatrix} O_{\beta\times(m-\alpha-\beta)} \\ X'_{\alpha\times(m-\alpha-\beta)} \end{pmatrix} \\ &= \begin{pmatrix} O_{(m-\alpha-\beta)\times\alpha} & Y'_{(m-\alpha-\beta)\times\beta} \end{pmatrix} \cdot P_{\alpha+\beta}^\beta \cdot \begin{pmatrix} O_{\beta\times(m-\alpha-\beta)} \\ \frac{1}{c_2} \cdot X'_{\alpha\times(m-\alpha-\beta)} \end{pmatrix} \\ &= \begin{pmatrix} O_{(m-\alpha-\beta)\times\alpha} & Y'_{(m-\alpha-\beta)\times\beta} \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{c_2} \cdot X'_{\alpha\times(m-\alpha-\beta)} \\ O_{\beta\times(m-\alpha-\beta)} \end{pmatrix} \\ &= O_{(m-\alpha-\beta)\times(m-\alpha-\beta)}, \end{aligned}$$

which implies that $\det(\tilde{M} - YN^{-1}X) = \det(\tilde{M})$. Consequently, we obtain $\det(M) = \det(N) \cdot \det(\tilde{M})$.

As for $\tilde{M}$, if $m-\alpha-\beta < \alpha+\beta$, then it is a zero matrix when $m-\alpha-\beta < \beta$, or a lower triangular matrix with zero main diagonal when $\beta \leq m-\alpha-\beta < \alpha$, or a 2-diagonal matrix with diagonals α and $-\beta$ otherwise. In all these cases, we have $\det(\tilde{M}) = 0$. If $m-\alpha-\beta \geq \alpha+\beta$, we recursively deal with $\tilde{M}$ as what we have done for M. So, if $\alpha+\beta \nmid m$, at last a block with dimension $t \times t$ where $t < \alpha+\beta$ will be left, which admits $\det(M) = 0$. □

Remark 2. Theorem 3 generalizes [GLG+17, Theorem 1], in which the case $p = 2$ is considered. In the above we give a totally different method with the one used in [GLG+17] to prove the general result, which is much easier to follow. Besides, from our proof, we can also derive the determiant of a 2-diagonal matrix when it is nonsingular, that is, $\det(M) = c_1^m$ when $\alpha = 0$ and otherwise,

$$\det(M) = \left[(-1)^{\alpha\beta} c_1^{\beta} c_2^{\alpha}\right]^{\frac{m}{\alpha+\beta}} \tag{7}$$

since $\det(P_{\alpha+\beta}^{\alpha}) = (-1)^{\alpha\beta}$. Although this formula is redundant in our study of MDS properties of rotation-add diffusion layers, it may have independent interest.

For 3-diagonal matrices, the situation is much more complicated, and it seems hopeless to derive a generic formula for its determinant and even a necessary and sufficient condition for it to be nonsingular. For example, when $\alpha_1 = 0$ and $\alpha_2 = \beta = 1$ in Definition 5, a 3-diagonal matrix is simply the well-known tri-diagonal matrix, whose determinant can be computed recursively and is related to the Lucas sequence. However, when $\alpha_1 = 0$ and $\alpha_2 + \beta = m$, we can derive a formula for the determinant.

Theorem 4. *Let M be an $m \times m$ 3-diagonal matrix over $\mathbb{F}_p$ with diagonals 0, α, $-(m-\alpha)$ and the corresponding constants $c_1,\ c_2,\ c_3$. Let $d = \gcd(\alpha, m)$. Then we have*

$$\det(M) = \left[c_1^{\frac{m}{d}} - (-1)^{\frac{m}{d}} c_2^{\frac{m-\alpha}{d}} c_3^{\frac{\alpha}{d}}\right]^d. \tag{8}$$

Proof. Note that M can be representated as

$$\begin{aligned} M &= c_1 \cdot I_m + c_2 \cdot U_m^{\alpha} + c_3 \cdot L_m^{m-\alpha} \\ &= c_1 \cdot I_m + \begin{pmatrix} c_2 \cdot I_{m-\alpha} & \\ & c_3 \cdot I_{\alpha} \end{pmatrix} \cdot P_m^{\alpha} \end{aligned}$$

where U_m, L_m and P_m are the upper shift matrix, lower shift matrix and cyclic permutation matrix of order m, respectively, as defined before. To compute $\det(M)$, we proceed via determining the characteristic polynomial of $D \cdot P_m^{\alpha}$ for a diagonal matrix D.

We consider a general D here, that is, $D = \mathrm{Diag}(\boldsymbol{z})$ for a vector $\boldsymbol{z} = (z_0, z_1, \ldots, z_{m-1}) \in \mathbb{F}_p^m$. Firstly, we assume α is coprime with m, namely, $d = 1$. In this case, m is the smallest number such that $(P_m^{\alpha})^m = I_m$ according to Proposition 2. Now we consider $(D \cdot P_m^{\alpha})^m$.

It is easy to check that

$$P_m^{\alpha} \cdot D = \mathrm{Diag}(\sigma(\boldsymbol{z})) \cdot P_m^{\alpha}$$

where $\sigma(\boldsymbol{z}) = (z_{\alpha}, z_{\alpha+1}, \ldots, z_{m-1}, z_0, \ldots, z_{\alpha-1})$, that is, σ is the permutation associated with the permutation matrix P_m^{α}. It follows that $P_m^{k\alpha} \cdot D = \mathrm{Diag}(\sigma^k(\boldsymbol{z})) \cdot P_m^{k\alpha}$ for any $1 \le k \le m-1$. Therefore, we have

$$\begin{aligned} (D \cdot P_m^{\alpha})^m &= (D \cdot P_m^{\alpha}) \cdot (D \cdot P_m^{\alpha}) \cdots (D \cdot P_m^{\alpha}) \\ &= D \cdot (P_m^{\alpha} D) \cdot P_m^{\alpha} \cdots (D \cdot P_m^{\alpha}) \\ &= D \cdot \mathrm{Diag}(\sigma(\boldsymbol{z})) \cdot P_m^{2\alpha} \cdot (D \cdot P_m^{\alpha}) \cdots (D \cdot P_m^{\alpha}) \\ &= D \cdot \mathrm{Diag}(\sigma(\boldsymbol{z})) \cdot \mathrm{Diag}(\sigma^2(\boldsymbol{z})) \cdot P_m^{3\alpha} \cdot (D \cdot P_m^{\alpha}) \cdots (D \cdot P_m^{\alpha}) \\ &= \cdots \ \cdots \\ &= \prod_{k=0}^{m-1} \mathrm{Diag}(\sigma^k(\boldsymbol{z})) \cdot P_m^{m\alpha} \\ &= \mathrm{Diag}\left(\bigcirc_{k=0}^{m-1} \sigma^k(\boldsymbol{z})\right) \cdot I_m, \end{aligned}$$

where "∘" represents Hadamard product (i.e., element-wise product) of vectors. Since $\gcd(\alpha, m) = 1$, we know that $\sigma^k(\boldsymbol{z})$ traverses all the m different circulant shifts of $\boldsymbol{z}$ for $0 \le k \le m-1$. Therefore, we have

$$\mathrm{Diag}\left(\bigcirc_{k=0}^{m-1}\sigma^k(\boldsymbol{z})\right) = \left(\prod_{k=0}^{m-1} z_k\right) \cdot I_m,$$

and it follows that $(D \cdot P_m^\alpha)^m = \left(\prod_{k=0}^{m-1} z_k\right) \cdot I_m$. This admits the characteristic polynomial of $D \cdot P_m^\alpha$, which is $\lambda^m - \prod_{k=0}^{m-1} z_k$.

With this observation, we can compute $\det(M)$ by putting

$$D = \begin{pmatrix} -c_2 \cdot I_{m-\alpha} & \\ & -c_3 \cdot I_\alpha \end{pmatrix}$$

and $\lambda = c_1$, getting that

$$\det(M) = \det\left(c_1 I_m - D \cdot P_m^\alpha\right) = c_1^m - (-1)^m c_2^{m-\alpha} c_3^\alpha.$$

In the case $\gcd(\alpha, m) = d > 1$, we can apply the tensor decomposition trick. More precisely, now we can represent M as

$$M = \left(c_1 \cdot I_{m/d} - D' \cdot P_{m/d}^{\alpha/d}\right) \otimes I_d,$$

where

$$D' = \begin{pmatrix} -c_2 \cdot I_{m/d-\alpha/d} & \\ & -c_3 \cdot I_{\alpha/d} \end{pmatrix}.$$

Hence, applying the formula (1), we generally obtain that

$$\det(M) = \det\left(c_1 \cdot I_{m/d} - D' \cdot P_{m/d}^{\alpha/d}\right)^d = \left[c_1^{\frac{m}{d}} - (-1)^{\frac{m}{d}} c_2^{\frac{m-\alpha}{d}} c_3^{\frac{\alpha}{d}}\right]^d.$$

□

Remark 3. Note that Theorem 4 holds for any $c_1,\ c_2,\ c_3 \in \mathbb{F}_p$. One can check that (8) coincides with (7) when $c_1 = 0$ due to the fact $(-1)^{\alpha\beta} = (-1)^{\alpha+\beta+\gcd(\alpha,\beta)}$. In their proof of [GLG+17, Theorem 5], the authors assert that a $b \times b$ 3-diagonal matrix of the form

$$\begin{pmatrix} \lambda & & \lambda+1 & & \\ & \lambda & & \ddots & \\ & & & & \lambda+1 \\ 1 & & \ddots & & \\ & \ddots & & & \\ & & 1 & & \lambda \end{pmatrix}$$

over $\mathbb{F}_2$ with diagonals 0, l and $-(b-l)$ has determinant $\lambda^b + (1+\lambda)^{b-l}$, without giving a proof or possible references. From our Theorem 4, we can see that this result is incorrect. The correct determinant should be $\left[\lambda^{b/d} + (1+\lambda)^{(b-l)/d}\right]^d$ where $d = \gcd(b, l)$. Similar kinds of determinants will also play an important role in our constructions of rotational-add MDS diffusion layers in the subsequent sections.

4 Characterizations of rotational-add MDS diffusion layers

In this section, based on the conditions on invertibility of 1-order submatrices established in the previous section, we aim to find a general form of the rotational-add diffusion layers that are potentially MDS for certain set $\mathcal{G} = \mathcal{I} \cup \mathcal{J}$ with $\mathcal{I} \cap \mathcal{J} = \emptyset$ and $|\mathcal{G}| = 5$. Firstly, we consider the case $\mathcal{G} = \mathcal{I}$. We find that there is only one possible form in this case. For this specific form, we divide further into several cases according to all possibilities of $\mathcal{I}$ and $\mathcal{J}$. It turns out that MDS matrices exist in every case. However, it is a challenging task to construct explicit classes of MDS matrices in all the cases. In the next section we will present some explicit constructions of MDS matrices for four special cases.

In the following discussions of this paper, we will always denote U and L to be the $m \times m$ upper and lower shift matrices, and P and I to be the $m \times m$ cyclic permutation matrix and identity matrix, respectively. Assume the first block-row of $L_{4,m}^{\mathcal{G}}$ is $(A,\ B,\ C,\ D)$ and $\mathcal{G} = \{g_1,\ g_2,\ g_3,\ g_4,\ g_5\}$ satisfies (6).

4.1 The case $|\mathcal{G}| = |\mathcal{I}|$

Note that $\mathcal{G} = \mathcal{I}$ means $L_{4,m}^{\mathcal{G}}(\boldsymbol{x}) = \sum_{i \in \mathcal{G}} (x_0, x_1, \ldots, x_{4m-1}) \lll i$. We study the conditions for $L_{4,m}^{\mathcal{G}}$ to be MDS under two distinct cases: $g_1 > 0$ and $g_1 = 0$. Firstly, we consider the situation for $g_1 > 0$.

Lemma 1. *Assume $g_1 > 0$. If $L_{4,m}^{\mathcal{G}}$ is MDS, then $m \nmid g_3,\ g_4,\ g_5$, which implies that $0 < g_1 < g_2 < m < g_3 < 2m < g_4 < 3m < g_5 < 4m$.*

Proof. If $m \mid g_5$, i.e., $g_5 = 3m$, we obtain $A = U^{g_1} + U^{g_2}$ is strictly upper triangular, which implies A is singular. If $m \nmid g_5$ but $m \mid g_4$, the matrix D takes the form of U^{g_5}, which is singular. The proof for $m \mid g_3$ follows analogously. □

From Lemma 1, we know that the submatrices C and D must be 2-diagonal matrices of the forms

$$C = U^{g_4 - 2m} + L^{2m - g_3} \text{ and } D = U^{g_5 - 3m} + L^{3m - g_4}.$$

If $L_{4,m}^{\mathcal{G}}$ is MDS, then from Theorem 3, we obtain $g_{i+1} - g_i \mid m$ for $i \in \{3,\ 4\}$. As $(g_5 - g_4) + (g_4 - g_3) = g_5 - g_3 > m$, there must exist one $i \in \{3,\ 4\}$ s.t. $g_{i+1} - g_i > m/2$, which implies $g_{i+1} - g_i = m$. Therefore, the remaining difference $g_{j+1} - g_j = s$ can be a proper divisor of m, or equal to m. The possible forms of the rotation offset $\mathcal{G}$ are given as follows:

1. $\{g_1,\ g_2,\ g_3,\ g_3 + m,\ g_3 + 2m\}$;
2. $\{g_1,\ g_2,\ g_3,\ g_4,\ g_4 + m\}$ where $g_4 = g_3 + s$ for a proper divisor s of m;
3. $\{g_1,\ g_2,\ g_3,\ g_3 + m,\ g_5\}$ where $g_5 = g_3 + m + s$ for a proper divisor s of m.

Proposition 11. *Let $\mathcal{G} = \{g_1,\ g_2,\ g_3,\ g_3 + m,\ g_3 + 2m\}$ with $0 < g_1 < g_2 < m < g_3 < 2m$. Then $L_{4,m}^{\mathcal{G}}$ cannot be MDS.*

Proof. In this case the submatrices B and D have the forms

$$B = L^{m-g_1} + L^{m-g_2} + U^{g_3 - m} \ \text{ and } \ D = L^{2m-g_3} + U^{g_3 - m}.$$

We obverse that $B - D = L^{m-g_1} + L^{m-g_2} - L^{2m-g_3}$ is strictly lower triangular, and thus singular. By Corollary 1, $L_{4,m}^{\mathcal{G}}$ cannot be MDS. □

Let $\mathcal{G} = \{g_1,\ g_2,\ g_3,\ g_4,\ g_4 + m\}$. By performing a left rotation of 3 blocks on $(L_{4,m}^{\mathcal{G}})^{\tau}$, we obtain $L_{4,m}^{\widetilde{\mathcal{G}}}$ with $\widetilde{\mathcal{G}} = \{\widetilde{g}_1,\ \widetilde{g}_2,\ \widetilde{g}_3,\ \widetilde{g}_3 + m,\ \widetilde{g}_5\} = \{m - g_2,\ m - g_1,\ 4m - g_4,\ 5m -$

$g_4,\ 5m - g_3\}$. By Proposition 5 and Proposition 6, the MDS properties of $L^{\mathcal{G}}_{4,m}$ and $L^{\widetilde{\mathcal{G}}}_{4,m}$ are equivalent. So, we can reduce the remaining two cases to a single case and focus on studying $\mathcal{G} = \{g_1,\ g_2,\ g_3,\ g_4,\ g_4 + m\}$. Our experimental results suggest that such MDS matrices do not exist. However, at this stage we have not come up with a theoretical proof of this non-existence. Since our goal in this paper is to construct explicit classes of rotational-add MDS diffusion layers, we leave it as an open problem for future research.

Next, we will explore all possible forms of $\mathcal{G}$ with $g_1 = 0$ such that $L^{\mathcal{G}}_{4,m}$ is MDS.

Proposition 12. *Let* $\mathcal{G} = \{0,\ g_2,\ g_3,\ g_4,\ g_5\}$ *with* $0 < g_2 < m \le g_3 < 2m \le g_4 < 3m \le g_5 < 4m$. *If* $L^{\mathcal{G}}_{4,m}$ *is MDS, then* $m \nmid g_3,\ g_4$, *which implies* $0 < g_2 < m < g_3 < 2m < g_4 < 3m \le g_5 < 4m$.

Proof. When $m \mid g_3$, i.e., $g_3 = m$, we find that for $C = U^{g_4 - 2m}$ to be nonsingular, it must hold $m \mid g_4$. Similarly, $m \mid g_4$ together with the non-singularity of D imply $m \mid g_5$. Then $L^{\mathcal{G}}_{4,m}$ can be expressed as $\mathrm{Circ}(A, B, I, I)$. However, in this case $B - D = B - I$ is strictly lower triangular, and thus singular. By Corollary 1, $L^{\mathcal{G}}_{4,m}$ cannot be MDS.

When $m \nmid g_3$ but $m \mid g_4$, we have $g_4 = 2m$, $g_5 = 3m$ as discussed above, and $L^{\mathcal{G}}_{4,m}$ has the form $\mathrm{Circ}(A, B, C, I)$. By Theorem 3, we obtain that $g_3 - g_2 \mid m$ since $B = U^{g_3 - m} + L^{m - g_2}$ is nonsingular. If $g_3 - g_2 < m$, i.e., $2m - g_3 > m - g_2$, we can see $A - C$ is a 2-diagonal matrix containing a zero row, thus is singular. If $g_3 - g_2 = m$, we have $B = P^{g_2}$ and then $B - D = B - I$ is a 3-diagonal matrix satisfying the conditions of Theorem 4. A simple computation shows $\det(B - D) = 0$ by Theorem 4, so $B - D$ is singular. Therefore, according to Corollary 1, $L^{\mathcal{G}}_{4,m}$ cannot be MDS. □

Proposition 13. *Let* $\mathcal{G} = \{0,\ g_2,\ g_3,\ g_4,\ g_5\}$ *with* $0 < g_2 < m < g_3 < 2m < g_4 < 3m \le g_5 < 4m$. *If* $L^{\mathcal{G}}_{4,m}$ *is MDS, then* $m \mid g_5$.

Proof. Assuming $m \nmid g_5$, we prove $L^{\mathcal{G}}_{4,m}$ is not MDS. The submatrices $B,\ C,\ D$ are all 2-diagonal matrices since $g_1 = 0$ and $m \nmid g_3,\ g_4$ according to Proposition 12. By Theorem 3, We obtain $g_{i+1} - g_i \mid m$ for $i \in \{2,\ 3,\ 4\}$. We claim there exist $i, j \in \{2,\ 3,\ 4\}$, $i \ne j$, s.t. $g_{i+1} - g_i = g_{j+1} - g_j = m$. Otherwise, $g_{i+1} - g_i \le m/2$ for at least two of $i \in \{2,\ 3,\ 4\}$ and then we get $g_5 - g_2 \le 2m$, which contradicts with $g_5 - g_2 > 2m$. So, at least two of $B,\ C,\ D$ are cyclic permutation matrix. WLOG, we assume B is a cyclic permutation matrix (i.e., $g_3 - g_2 = m$) since otherwise, we can consider $L^{\mathcal{G}'}_{4,m}$, the transpose of $L^{\mathcal{G}}_{4,m}$, with $\mathcal{G}' = \{0,\ 4m - g_5,\ 4m - g_4,\ 4m - g_3,\ 4m - g_2\}$. Now we have $B = P^{g_2}$.

Denote by $\alpha = g_4 - 2m$, $\beta = g_5 - 3m$. Then we have $g_2 \ge \alpha \ge \beta$. When D is a cyclic permutation matrix (i.e., $\alpha = \beta$), we have $D = P^{\beta}$. It follows that $B - D = P^{\beta}(P^{g_2 - \beta} - I)$, which is obviously singular applying Theorem 4 again (or one can also directly figure this out from the fact that 1 is an eigenvalue of any cyclic permutation matrix).

When C is a cyclic permutation matrix, we have $g_2 = \alpha \ge \beta$ and now $C = B = P^{\alpha}$ and $D = U^{\beta} + L^{m - \alpha}$. Consider the 2-order submatrix

$$V = \begin{pmatrix} C & D \\ B & C \end{pmatrix} = \begin{pmatrix} P^{\alpha} & U^{\beta} + L^{m-\alpha} \\ P^{\alpha} & P^{\alpha} \end{pmatrix}.$$

By Proposition 3, we obtain

$$\begin{aligned} \det(V) &= \det(P^{2\alpha} - (U^{\beta} + L^{m-\alpha}) \cdot P^{\alpha}) \\ &= \det((U^{\alpha} + L^{m-\alpha}) - (U^{\beta} + L^{m-\alpha})) \cdot \det(P^{\alpha}) \\ &= \det(U^{\alpha} - U^{\beta}) \cdot \det(P^{\alpha}) \\ &= 0 \end{aligned}$$

since $U^{\alpha} - U^{\beta}$ is strictly upper triangular.

Finally, we know $L^{\mathcal{G}}_{4,m}$ cannot be MDS if $m \nmid g_5$. □

Summarizing the above analysis, finally we find that when $|\mathcal{G}| = |\mathcal{I}| = 5$ and $g_1 = 0$, there is only one candidate class of $L^{\mathcal{G}}_{4,m}$ that is potentially MDS, showing in the following proposition.

Proposition 14. *Let $\mathcal{G} = \{0,\ g_2,\ g_3,\ g_4,\ 3m\}$ with $0 < g_2 < m < g_3 < 2m < g_4 < 3m$. If $L^{\mathcal{G}}_{4,m}$ is MDS, then $g_3 = g_2 + m,\ g_4 = g_2 + 2m$.*

Proof. As $g_2 < m$ and $g_4 > 2m$, we have $g_4 - g_2 > m$. Considering a proper divisor of m is no bigger than $m/2$, we find at most one of $g_3 - g_2$ and $g_4 - g_3$ is a proper divisor of m. Denote by $\alpha = g_3 - m$, $\beta = g_4 - 2m$. By Theorem 3, there are only 3 possibilities for $\mathcal{G}$:

1. $\{0,\ g_2,\ g_2 + m,\ g_4,\ 3m\}$ with $s = g_4 - (g_2 + m) = m - g_2 + \beta$ being a proper divisor of m. In this case, $g_2 = \alpha$, $g_2 - \beta \geq m/2$, and $A = I + U^{\alpha}$, $B = U^{\alpha} + L^{m-\alpha}$, $C = U^{\beta} + L^{m-\alpha}$, $D = I + L^{m-\beta}$;

2. $\{0,\ g_2,\ g_3,\ g_3 + m,\ 3m\}$ with $s = g_3 - g_2 = m - g_2 + \alpha$ being a proper divisor of m. In this case, $g_2 - \alpha \geq m/2$, and $A = I + U^{g_2}$, $B = U^{\alpha} + L^{m-g_2}$, $C = U^{\alpha} + L^{m-\alpha}$, $D = I + L^{m-\alpha}$;

3. $\{0,\ g_2,\ g_2 + m,\ g_2 + 2m,\ 3m\}$.

For Case 1, we prove the submatrix

$$T = \begin{pmatrix} A & D \\ C & B \end{pmatrix} = \begin{pmatrix} I + U^{\alpha} & I + L^{m-\beta} \\ U^{\beta} + L^{m-\alpha} & U^{\alpha} + L^{m-\alpha} \end{pmatrix}$$

is singular. An elementary column transformation of T admits

$$T' = \begin{pmatrix} A - D & D \\ C - B & B \end{pmatrix} = \begin{pmatrix} U^{\alpha} - L^{m-\beta} & I + L^{m-\beta} \\ U^{\beta} - U^{\alpha} & U^{\alpha} + L^{m-\alpha} \end{pmatrix}.$$

Obviously, the $(m-\alpha)$-th to $(m-\beta-1)$-th rows of $A - D$ are all $\mathbf{0}$. Let $t = \alpha - \beta$. Then we have $m - \alpha \leq t \leq m - \beta - 1$ since $\alpha - \beta \geq m/2$ and $\alpha \leq m - 1$. It follows that the t-th row of D is $\boldsymbol{e}_t$. Consider the $(m - \alpha + t)$-th, i.e., the $(m-\beta)$-th rows of $C - B$ and B. We can observe that they are $\mathbf{0}$ and $\boldsymbol{e}_t$, respectively. Therefore, T' contains two identical rows, thus it must be singular.

Besides, one can figure out that for any $L^{\mathcal{G}'}_{4,m}$ in Case 2, there exists an $L^{\mathcal{G}}_{4,m}$ in Case 1 that is equivalent to it according to Proposition 5 and Proposition 6. As a result, any matrix in Case 2 cannot be MDS as well. So, the only possibility is Case 3. ☐

4.2 The special case $\mathcal{G} = \{0,\ g_2,\ g_2 + m,\ g_2 + 2m,\ 3m\}$

The analysis in Section 4.1 can also be extended to the general situation, that is, $\mathcal{G} = \mathcal{I} \cup \mathcal{J}$ with a nonempty $\mathcal{J}$. That will help us to establish more non-existence results on rotational-add MDS diffusion layers and get full classifications of them. However, the discussions will be rather complicated since we should consider all possible partitions of $\mathcal{G}$ and the corresponding distributions of $g_1, \ldots, g_5$. Since the main purpose of the current paper is to construct some explicit classes of lightest rotational-add MDS diffusion layers, we leave this full classification task as a further work. At present, we only focus on classification of MDS diffusion layers in the sole remaining type of $L^{\mathcal{G}}_{4,m}$ that is specified in the case $\mathcal{G} = \mathcal{I}$ in Section 4.1.

For $\mathcal{G} = \{0,\ g_2,\ g_2 + m,\ g_2 + 2m,\ 3m\}$ where $0 < g_2 < m$, we consider the partition $\mathcal{G} = \mathcal{I} \cup \mathcal{J}$ where $|\mathcal{I}| \leq |\mathcal{G}| = 5$. In total we have $\sum_{k=0}^{5} \binom{5}{k} = 32$ possibilities. However, considering the relation between $L^{\mathcal{G}}_{4,m}$ and $-L^{\mathcal{G}}_{4,m}$, only a half needs to be taken into account, namely, 16 cases according as $|\mathcal{I}| = 5$, $|\mathcal{I}| = 4$ and $|\mathcal{I}| = 3$.

Figure 1 shows forms of the first block-row of $L_{4,m}^{\mathcal{G}}$ in these 16 different cases. Note that in Figure 1, we use a black line to represent a diagonal with entries 1, while a blue one to represent a diagonal with entries -1. Experimental results given in Appendix A demonstrate existence of MDS matrices in all these 16 cases.

Carefully examining matrices in all these 16 cases, we observe that $L_{4,m}^{\mathcal{G}}$ fall into 8 different classes in consideration of equivalence between them (the equivalence comes from transpose of matrices or circulant permutations of block-rows). Different forms of these 8 classes of $L_{4,m}^{\mathcal{G}}$ are listed below:

1. $\mathrm{Circ}(A, B, B, B - A + 2I)$ for Case 1;
2. $\mathrm{Circ}(A, B, B, B - A)$ for Case 2 (equivalent to Case 6);
3. $\mathrm{Circ}(A, B, -2(A - I) + B, A - B)$ for Case 3 (equivalent to Case 5) and Case 11 (equivalent to Case 14);
4. $\mathrm{Circ}(A, B, -B, A + B)$ for Case 4 and Case 12;
5. $\mathrm{Circ}(A, B, -2(A + I) - B, -(A + B))$ for Case 7 (equivalent to Case 16);
6. $\mathrm{Circ}(A, B, -B, A + B + 2I)$ for Case 8 (equivalent to Case 15);
7. $\mathrm{Circ}(A, B, -2(A - I) - B, -(A + B))$ for Case 9 (equivalent to Case 13);
8. $\mathrm{Circ}(A, B, B, B - A - 2I)$ for Case 10.

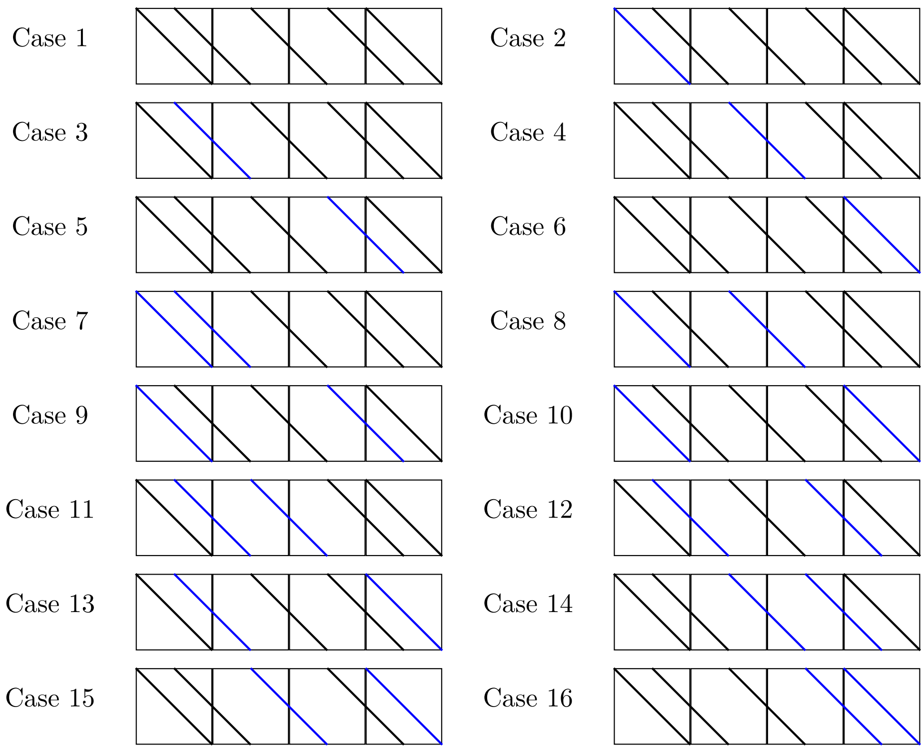

Figure 1: 16 possible cases for distributions of ± 1 in $L_{4,m}^{\mathcal{G}}$

5 Direct constructions of rotational-add MDS diffusion layers

In this section, we focus on explicit constructions of rotation-add MDS diffusion layers for $\mathcal{G} = \{0,\ g_2,\ g_2+m,\ g_2+2m, 3m\}$. We investigate necessary and sufficient conditions for matrices of the forms $L_{4,m}^{\mathcal{G}} = \mathrm{Circ}(A, B, B, B-A)$ (corresponding to Case 2 and Case 6 in Figure 1) and $L_{4,m}^{\mathcal{G}} = \mathrm{Circ}(A, B, -B, A+B)$ (corresponding to Case 4 and Case 12 in Figure 1) to be MDS. For all these four cases, it turns out that we can fully characterize the MDS properties of $L_{4,m}^{\mathcal{G}}$ and obtain some explicit constructions of MDS matrices under certain special choices of parameters.

5.1 Rotational-add MDS diffusion layers of the form $\mathrm{Circ}(\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{B}, \boldsymbol{B}-\boldsymbol{A})$

We study $L_{4,m}^{\mathcal{G}} = \mathrm{Circ}(A, B, B, B-A)$ at first. Actually, we can characterize MDS property of $L_{4,m}^{\mathcal{G}}$ by nonzeroness of a list of determinants. To make these conditions more tractable, we further characterize them in terms of the eigenvalues of BA^{-1} for arbitrary values of the parameters m, g_2 and p. Based on this general result, we proceed to explicitly construct a class of MDS matrices for some special choices of parameters.

In the following, we always denote BA^{-1} by W.

Proposition 15. *Let $L_{4,m}^{\mathcal{G}} = \mathrm{Circ}(A, B, B, B-A)$ with $A,\ B,\ B-A \in GL(m, \mathbb{F}_p)$. Then $L_{4,m}^{\mathcal{G}}$ is an MDS matrix if and only if determinants of matrices in the following list are all nonzero:*

1. $W + I$;
2. $2W - I$;
3. $W - 2I$;
4. $W^2 - W - I$;
5. $W^2 - W + I$;
6. $W^2 - 3W + I$;
7. $3(W^2 - 2W + 2I)$;
8. $W^3 - W^2 - I$;
9. $2W^3 - 5W^2 + 3W + I$;
10. $W^3 - 4W^2 + 3W - I$;
11. $W^3 - 4W^2 + 6W - I$.

Proof. Let $D = I_4 \otimes A^{-1}$ be a block diagonal matrix. Then we have

$$L_{4,m}^{\mathcal{G}} \cdot D = \begin{pmatrix} I & W & W & W-I \\ W-I & I & W & W \\ W & W-I & I & W \\ W & W & W-I & I \end{pmatrix} := L_{4,m}^{\mathcal{G}\star}.$$

The branch numbers of $L_{4,m}^{\mathcal{G}}$ and $L_{4,m}^{\mathcal{G}\star}$ are equal according to Proposition 7. Consequently, we need only to derive the necessary and sufficient conditions for $L_{4,m}^{\mathcal{G}\star}$ to be MDS. Noting that I, W and $W-I$ are all nonsingular and pairwise commutative, we can directly compute determinants of all the submatrices appearing in Proposition 10 based on Theorem 1. It is worth noting that the determinant of $L_{4,m}^{\mathcal{G}\star}$ itself is given by

$$\det(L_{4,m}^{\mathcal{G}\star}) = \det\left(3 \cdot (W^4 - 4W^3 + 6W^2 - 4W)\right),$$

which admits the following factorization:

$$\begin{aligned}\det(L_{4,m}^{\mathcal{G}\star}) &= \det\left(3 \cdot W \cdot (W - 2I) \cdot (W^2 - 2W + 2I)\right)\\ &= 3^m \cdot \det(W) \cdot \det(W - 2I) \cdot \det(W^2 - 2W + 2I).\end{aligned}$$

Considering this fact, we can get all these determinants that should be checked for $L_{4,m}^{\mathcal{G}\star}$ to be MDS. □

From Proposition 15, it is obvious that no MDS matrix exists in this form when $p = 3$.

The determinant of a matrix is nonzero is equivalent to the eigenvalues of it are all nonzero. Let λ be an eigenvalue of W. Of course, λ^k is an eigenvalues of W^k. Then the determinant conditions for W in Proposition 15 can be equivalently transformed to conditions on λ. Note that the characteristic polynomial of W is given by

$$\det(\lambda I - W) = \det(\lambda A - B) \cdot \det(A^{-1}).$$

So, it suffices to compute $\det(\lambda A - B)$ since $\det(A^{-1}) = \pm 1$. Fortunately, $\lambda A - B$ is a 3-diagonal matrix with diagonals 0, g_2 and $-(m - g_2)$, whose determinant is an immediate consequence of Theorem 4. As a result, for $L_{4,m}^{\mathcal{G}} = \mathrm{Circ}(A, B, B, B - A)$, we can obtain the necessary and sufficient conditions on the parameters m, g_2 and p for it to be MDS.

Theorem 5. *Let $L_{4,m}^{\mathcal{G}} = \mathrm{Circ}(A, B, B, B - A)$ with $A,\ B,\ B - A \in GL(m, \mathbb{F}_p)$ having the forms in Case 2 of Figure 1. Assume $p \neq 3$ and $d = \gcd(g_2,\ m)$. Let*

$$f(\lambda) = (-\lambda)^{\frac{m}{d}} - (1 - \lambda)^{\frac{m-g_2}{d}}. \tag{9}$$

Then $L_{4,m}^{\mathcal{G}}$ is MDS if and only if each of the following polynomials is coprime to $f(\lambda)$:

1. $\lambda + 1$;
2. $2\lambda - 1$;
3. $\lambda - 2$;
4. $\lambda^2 - \lambda - 1$;
5. $\lambda^2 - \lambda + 1$;
6. $\lambda^2 - 3\lambda + 1$;
7. $\lambda^2 - 2\lambda + 2$;
8. $\lambda^3 - \lambda^2 - 1$;
9. $2\lambda^3 - 5\lambda^2 + 3\lambda + 1$;
10. $\lambda^3 - 4\lambda^2 + 3\lambda - 1$;
11. $\lambda^3 - 4\lambda^2 + 6\lambda - 1$.

Proof. For Case 2 in Figure 1, we have

$$A = -I + U^{g_2},\ B = U^{g_2} + L^{m-g_2}.$$

Consequently, $\lambda A - B$ is a 3-diagonal matrix with diagonals $0,\ g_2,\ -(m-g_2)$, and constants $c_1 = -\lambda$, $c_2 = \lambda - 1$, $c_3 = -1$ (see notations in Definition 5). By Theorem 4, we can obtain that

$$\det(\lambda A - B) = \left[(-\lambda)^{\frac{m}{d}} - (-1)^{\frac{m-g_2}{d}}(\lambda - 1)^{\frac{m-g_2}{d}}\right]^d = f(\lambda)^d.$$

As a result, the matrices corresponding to Case 2 of Figure 1 are MDS if and only if $f(\lambda)$ is coprime to each of the polynomials derived correspondingly from each of the determinant conditions listed in Proposition 15. □

Theorem 5 reduces the conditions for $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, B, B - A)$ being MDS to constraints on the eigenvalues of W. To verify these conditions, according to Proposition 4, we can compute $\mathrm{Res}(f,\ g)$ for each polynomial g listed in Theorem 5 given the parameters m, g_2 and p. This verification process is very efficient, which will help us to find MDS matrices easily for symmetric cipher designs. In fact, since all these polynomials have integer coefficients, $\mathrm{Res}(f,\ g)$ is an integer for every g, which should be nonzero modulo p. Therefore, if p is much larger than m (e.g., $p = 65537$ and $m = 4$) and $\deg(f)$ is small, $L^{\mathcal{G}}_{4,m}$ can almost definitely be MDS since $\mathrm{Res}(f,\ g)$ may be a nonzero integer smaller than p for every g.

Although it seems hopeless to theoretically derive a generic characterization on m, g_2 and p for $L^{\mathcal{G}}_{4,m}$ to be MDS, some necessary conditions are easy to obtain. For example, there are 3 polynomials of degree 1 in Theorem 5, namely, $\lambda + 1$, $2\lambda - 1$ and $\lambda - 2$. They are coprime to $f(\lambda)$ if and only if $p = 2$ or $p > 2$ and $f(-1)$, $f(1/2)$ and $f(2)$ are all nonzero (note here that $1/2$ is the inverse of 2 modulo p). This admits the necessary conditions

$$p \nmid 2^{\frac{m-g_2}{d}} \text{ (i.e., } p \neq 2),\ p \nmid \left(2^{\frac{g_2}{d}} - (-1)^{\frac{m}{d}}\right) \text{ and } p \nmid \left(2^{\frac{m}{d}} - (-1)^{\frac{g_2}{d}}\right).$$

Note that if $f(\lambda)$ has a low degree, we can compute $\mathrm{Res}(f,\ g)$ for each g listed in Theorem 5 by such math software as SageMath easily. Low-degree f can be achieved by imposing certain relations between m and g_2, which will yield explicit classes of MDS matrices of the form $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, B, B - A)$. For example, we present the following class of MDS matrices.

Corollary 2. *Let $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, B, B - A)$ with $A,\ B,\ B - A \in GL(m, \mathbb{F}_p)$ having the forms in Case 2 of Figure 1. Assume m is even and $g_2 = m/2$. Then $L^{\mathcal{G}}_{4,m}$ is MDS if and only if $p \neq 2,\ 3,\ 5$.*

Proof. From (9), we have in this case,

$$f(\lambda) = \lambda^2 + \lambda - 1.$$

We compute $\mathrm{Res}(f,\ g)$ for each g listed in Theorem 5, and in turn the results are

$$-1,\ -1,\ 5,\ -4,\ 4,\ -4,\ 9,\ 9,\ -36,\ 9,\ -36.$$

These resultants must be nonzero over $\mathbb{F}_p$, which implies $p \neq 2,\ 3,\ 5$. □

Remark 4. As Case 2 and Case 6 in Figure 1 are equivalent, we can derive a class of construction for Case 6 immediately from Case 2. When m is even and $g_2 = m/2$, the matrix $L^{\mathcal{G}}_{4,m}$ corresponding to Case 6 in Figure 1 is MDS if and only if $p \neq 2,\ 3,\ 5$ as well.

5.2 Rotational-add MDS diffusion layers of form $\mathrm{Circ}(\boldsymbol{A}, \boldsymbol{B}, -\boldsymbol{B}, \boldsymbol{A}+\boldsymbol{B})$

In this subsection, we follow the similar analysis as above for $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$, which is in Case 4 and Case 12 of Figure 1. We also characterize MDS property of $L^{\mathcal{G}}_{4,m}$ by constraints on determinants and eigenvalues of W. We will provide some appropriate choices for the parameters to construct MDS matrices in these two cases along with an impracticable example.

Proposition 16. *Let $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$ with $A,\ B,\ A+B \in GL(m, \mathbb{F}_p)$. Then $L^{\mathcal{G}}_{4,m}$ is an MDS matrix if and only if determinants of the following matrices are all nonzero:*

1. $2W + I$;
2. $W + 2I$;
3. $W^2 - I$;
4. $W^2 + W - I$;
5. $W^2 + W + I$;
6. $W^2 + 3W + I$;
7. $3 \cdot (W^2 + 2W + 2I)$;
8. $W^3 + W^2 + I$;
9. $W^3 + 4W^2 + 3W + I$;
10. $W^3 + 4W^2 + 6W + I$;
11. $2W^3 + 5W^2 + 3W - I$.

Proof. The proof is similar to the proof of Proposition 15. Consider the determinant of $L^{\mathcal{G}\star}_{4,m} = L^{\mathcal{G}}_{4,m} \cdot (I_4 \otimes A^{-1})$, which is given by

$$\det(L^{\mathcal{G}\star}_{4,m}) = \det(3 \cdot (W^4 + 4W^3 + 6W^2 + 4W)).$$

It admits the following factorization:

$$\det(L^{\mathcal{G}\star}_{4,m}) = 3^m \cdot \det(W) \cdot \det(W + 2I) \cdot \det(W^2 + 2W + 2I).$$

The detailed proof is omitted here for brevity and left to the interested readers. □

From Proposition 16, it is obvious that no MDS matrix exists in this form when $p = 3$. Using the same method as in Section 5.1, we convert Proposition 16 into constraints on the eigenvalue λ of W.

Theorem 6. *Let $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$ with $A,\ B,\ A+B \in GL(m, \mathbb{F}_p)$ having the forms in Case 4 (or Case 12) of Figure 1. Assume $p \neq 3$ and $d = \gcd(g_2,\ m)$. Let*

$$f_1(\lambda) = \lambda^{\frac{m}{d}} - (-\lambda - 1)^{\frac{m-g_2}{d}} \tag{10}$$

and

$$f_2(\lambda) = \lambda^{\frac{m}{d}} - (-1)^{\frac{g_2}{d}} (\lambda + 1)^{\frac{m-g_2}{d}}. \tag{11}$$

Then $L^{\mathcal{G}}_{4,m}$ in Case 4 (or Case 12) of Figure 1 is MDS if and only if each of the following polynomials is coprime to $f_1(\lambda)$ (or $f_2(\lambda)$):

1. $2\lambda+1$;
2. $\lambda+2$;
3. λ^2-1;
4. $\lambda^2+\lambda-1$;
5. $\lambda^2+\lambda+1$;
6. $\lambda^2+2\lambda+2$;
7. $\lambda^2+3\lambda+1$;
8. $\lambda^3+\lambda^2+1$;
9. $\lambda^3+4\lambda^2+3\lambda+1$;
10. $\lambda^3+4\lambda^2+6\lambda+1$;
11. $2\lambda^3+5\lambda^2+3\lambda-1$.

Proof. When $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$, the submatrices A and B admit two possible forms corresponding to Case 4 and Case 12 in Figure 1, respectively. For Case 4, we have

$$A = I + U^{g_2},\ B = -U^{g_2} + L^{m-g_2}.$$

Thus $\lambda A - B$ is a 3-diagonal matrix with diagonals $0,\ g_2,\ -(m-g_2)$, and constants $c_1=\lambda$, $c_2=\lambda+1$, $c_3=-1$ (see notations in Definition 5). By Theorem 4, we can then obtain

$$\det(\lambda A - B) = \left[\lambda^{\frac{m}{d}} - (-1)^{\frac{m+g_2}{d}}(\lambda+1)^{\frac{m-g_2}{d}}\right]^d = f_1(\lambda)^d.$$

For Case 12, we find that

$$A = I - U^{g_2},\ B = U^{g_2} - L^{m-g_2}.$$

Thus $\lambda A - B$ is a 3-diagonal matrix with diagonals $0,\ g_2,\ -(m-g_2)$, and constants $c_1=\lambda$, $c_2=-(\lambda+1)$, $c_3=1$. We can obtain

$$\det(\lambda A - B) = \left[\lambda^{\frac{m}{d}} - (-1)^{\frac{m}{d}}(-\lambda-1)^{\frac{m-g_2}{d}}\right]^d = f_2(\lambda)^d.$$

As a result, the matrices corresponding to Case 4 (or Case 12) of Figure 1 are MDS if and only if $f_1(\lambda)$ (or $f_2(\lambda)$) is coprime to every polynomial derived correspondingly from each of the determinant conditions listed in Proposition 16. □

Remark 5. Note that Theorem 6 and Theorem 5 coincide when $p=2$. In this case, the necessary and sufficient conditions for $L^{\mathcal{G}}_{4,m}$ to be MDS are simplified to

$$\gcd(f(\lambda),\ \lambda^2+\lambda+1) = 1, \tag{12}$$
$$\gcd(f(\lambda),\ \lambda^3+\lambda+1) = 1, \tag{13}$$
$$\gcd(f(\lambda),\ \lambda^3+\lambda^2+1) = 1, \tag{14}$$

where $f(\lambda) = \lambda^{m/d} + (\lambda+1)^{(m-g_2)/d}$, $d=\gcd(g_2,m)$. It is easy to see (12) is equivalent to $\gcd(\lambda^{m/d} + \lambda^{2(m-g_2)/d},\ \lambda^2+\lambda+1) = \gcd(1+\lambda^{(m-2g_2)/d},\ \lambda^2+\lambda+1) = 1$, which can be reduced to

$$\frac{g_2}{d} \not\equiv \frac{2m}{d} \pmod 3 \tag{15}$$

since $\lambda^2+\lambda+1$ is a primitive polynomial over $\mathbb{F}_2$ with period 3. Similarly, (13) is equivalent to

$$\frac{g_2}{d} \not\equiv \frac{3m}{d} \pmod 7 \tag{16}$$

since $\lambda^3+\lambda+1$ is a primitive polynomial over $\mathbb{F}_2$ with period 7. As for (14), it is equivalent to $\gcd(f^*(\lambda),\ \lambda^3+\lambda+1) = 1$ where $f^*(\lambda) = \lambda^{m/d} \cdot f(1/\lambda)$ is the reciprocal polynomial of f. Then the condition can be reduced to

$$\frac{g_2}{d} \not\equiv \frac{5m}{d} \pmod 7. \tag{17}$$

We point out that [GLG+17, Theorem 5 and Theorem 6] characterizing MDS property of $L^{\mathcal{G}}_{4,m}$ in the binary case, are both incorrect. (15), (16) and (17) constitute the correct conditions.

Following the same approach as in Theorem 5, we can obtain constraints on p for any given m and g_2 via computing resultants. We can also derive some explicit constructions for specific parameters. Firstly, we present a non-existence result on MDS matrices in a certain class of $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$.

Corollary 3. *Let $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$ with $A,\ B,\ A+B \in GL(m, \mathbb{F}_p)$ having the forms in Case 4 or Case 12 of Figure 1. When m is even and $g_2 = m/2$, $L^{\mathcal{G}}_{4,m}$ cannot be MDS.*

Proof. When m is even and $g_2 = m/2$, by Theorem 6, we find that in (10) and (11),

$$f_1(\lambda) = f_2(\lambda) = \lambda^2 + \lambda + 1,$$

which is exactly equal to the fifth polynomial listed in Theorem 6, thereby implying that $L^{\mathcal{G}}_{4,m}$ cannot be an MDS matrix. □

For $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$ with $\mathcal{G} = \{0,\ g_2,\ g_2+m,\ g_2+2m,\ 3m\}$ and $g_2 < \frac{m}{2}$, there must exist an equivalent form $L^{\mathcal{G}'}_{4,m}$ where $\mathcal{G}' = \{0,\ m-g_2,\ 2m-g_2,\ 3m-g_2,\ 3m\}$ with $m - g_2 > \frac{m}{2}$. It is obtained by performing a left rotation of one block on $(L^{\mathcal{G}}_{4,m})^\tau$, in the form of $\mathrm{Circ}(A', B', -B', A'+B')$ as well. By Propositions 5 and Proposition 6, the MDS properties of $L^{\mathcal{G}}_{4,m}$ and $L^{\mathcal{G}'}_{4,m}$ are equivalent. Therefore, when dealing with $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$, it suffices to consider the situation that $g_2 > \frac{m}{2}$.

Corollary 4. *Let $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$ with $A,\ B,\ A+B \in GL(m, \mathbb{F}_p)$ having the forms in Case 4 of Figure 1. Assume $3 \mid m$ and $g_2 = \frac{2m}{3}$. Then $L^{\mathcal{G}}_{4,m}$ is MDS if and only if $p \neq 2,\ 3,\ 11,\ 13,\ 47$.*

Proof. From (10), we have in this case,

$$f_1(\lambda) = \lambda^3 + \lambda + 1.$$

We compute $\mathrm{Res}(f,\ g)$ for each g listed in Theorem 6 and in turn the results are

$$-3,\ 9,\ -3,\ -9,\ 3,\ 13,\ -11,\ 3,\ 24,\ -141,\ -141.$$

These resultants must be nonzero over $\mathbb{F}_p$, which implies $p \neq 2,\ 3,\ 11,\ 13,\ 47$. □

Corollary 5. *Let $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A+B)$ with $A,\ B,\ A+B \in GL(m, \mathbb{F}_p)$ having the forms in Case 12 of Figure 1. Assume $3 \mid m$ and $g_2 = \frac{2m}{3}$. Then $L^{\mathcal{G}}_{4,m}$ is MDS if and only if $p \neq 2,\ 3,\ 5,\ 7,\ 11,\ 61,\ 101$.*

Proof. From (11), we have in this case

$$f_2(\lambda) = \lambda^3 - \lambda - 1.$$

We compute $\mathrm{Res}(f,\ g)$ for each g listed in Theorem 6 and in turn the results are

$$5,\ 7,\ 1,\ 5,\ 1,\ 5,\ 11,\ 11,\ 8,\ 101,\ 61.$$

These resultants must be nonzero over $\mathbb{F}_p$, which implies $p \neq 2,\ 3,\ 5,\ 7,\ 11,\ 61,\ 101.$ □

Examining the explicit constructions of $L^{\mathcal{G}}_{4,m}$ that are MDS given in Corollary 2, Corollary 4 and Corollary 5, we find that the key point is the determinant of the corresponding matrix $\lambda A - B$ has the form $f(\lambda)^d$ with a low-degree polynomial $f(\lambda)$, which enables to derive conditions on p by explicitly computing resultants. Note that the corresponding $f(\lambda)$ in both Theorem 5 and Theorem 6 has degree m/d. Therefore, for any m and a divisor d such that $t = m/d$ is small, we can choose g_2 from the set

$$\mathcal{G}_2 = \{0 < g_2 < m \mid \gcd(m, g_2) = d\} = \left\{ k\frac{m}{t} \;\middle|\; 0 < k < t,\ \ \gcd(k,t) = 1 \right\}.$$

Then the corresponding $f(\lambda)$ has a low degree and as a result, a variety of constructions can be obtained in the form of $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, B, B - A)$ or $L^{\mathcal{G}}_{4,m} = \mathrm{Circ}(A, B, -B, A + B)$ following the aforementioned methodology.

In Appendix B, for an integer t from 2 up to 16 under the condition $t \mid m$ for any m, we list all g_2 in the set $\mathcal{G}_2$ and the corresponding conditions for $L^{\mathcal{G}}_{4,m}$ to be MDS (i.e., constraints on the prime p) in Case 2, Case 4 and Case 12 in Figure 1, respectively. In practical applications, one can flexibly design rotational-add MDS diffusion layers of various types by selecting appropriate parameters m according to the specific requirements of the symmetric ciphers.

5.3 The inverses of proposed rotational-add MDS diffusion layers

For the three example classes of rotational-add MDS diffusion layers presented in Corollary 2, Corollary 4 and Corollary 5, we can also derive their inverses. In fact, since $L^{\mathcal{G}}_{4,m}$ is itself an $mn \times mn$ circulant matrix over $\mathbb{F}_p$, we can compute $(L^{\mathcal{G}}_{4,m})^{-1}$ by computing the inverse of its associated polynomial modulo $x^{4m} - 1$. This is easier to tackle than directly computing inverse matrix of $L^{\mathcal{G}}_{4,m}$.

Recall that in these three cases, the corresponding associated polynomials of $L^{\mathcal{G}}_{4,m}$, respectively, are:

1. Corollary 2: $-1 + x^{g_2} + x^{g_2+m} + x^{g_2+2m} + x^{3m}$;
2. Corollary 4: $1 + x^{g_2} - x^{g_2+m} + x^{g_2+2m} + x^{3m}$;
3. Corollary 5: $1 - x^{g_2} + x^{g_2+m} - x^{g_2+2m} + x^{3m}$.

In Proposition 17, we directly give the inverse polynomials of these three polynomials under the corresponding conditions of Corollary 2, Corollary 4 and Corollary 5, respectively. Correctness of them can be checked simply and will be omitted.

Proposition 17. *1. Corollary 2:*

$$\frac{1}{15} \cdot \left(-4 - 2x + 4x^2 + 2x^3 + x^4 + 3x^5 - x^6 + 2x^7\right) \circ x^{\frac{m}{2}};$$

2. Corollary 4:

$$\frac{1}{117} \cdot \big(40 + 7x - 29x^2 + 31x^3 + 22x^4 - 2x^5 - 14x^6 - 20x^7 + 16x^8 - 5x^9 + 4x^{10} - 11x^{11}\big) \circ x^{\frac{m}{3}};$$

3. Corollary 5:

$$\frac{1}{105}\cdot\big(36+23x+39x^2+57x^3+16x^4+18x^5-6x^6 \\ +2x^7-24x^8-27x^9-26x^{10}-3x^{11}\big)\circ x^{\frac{m}{3}}.$$

Note here that "$\circ$" represents composition of polynomials, and all the fractions are taken over the corresponding field $\mathbb{F}_p$ (recall the corresponding constraints on p in each case).

We explain the idea for computing the inverse polynomials by one simple example. For Corollary 2, note that the associated polynomial $f(x)$ can be factorized as

$$\begin{aligned} f(x) &= -1+x^{g_2}+x^{g_2+m}+x^{g_2+2m}+x^{3m} \\ &= (-1+x^{g_2}+x^m)\cdot(1+x^m+x^{2m}) \\ &:= f_1(x)\cdot f_2(x). \end{aligned}$$

Then, we can compute $f^{-1}(x) \pmod{x^{4m}-1}$ by computing $f_1^{-1}(x) \pmod{x^{4m}-1}$ and $f_2^{-1}(x) \pmod{x^{4m}-1}$ separately, then we have $f(x)^{-1}=f_1(x)^{-1}\cdot f_2(x)^{-1}$. To compute $f_2^{-1}(x)$, we need only to compute $(1+x+x^2)^{-1} \pmod{x^4-1}$; to compute $f_1^{-1}(x)$, we assume $m=2e$ and then $f_2(x)=-1+x^e+x^{2e}$, so we can transform the problem to computing $(-1+x+x^2)^{-1} \pmod{x^8-1}$. In both cases we can apply the extended Euclidean algorithm directly. For Corollary 4 and Corollary 5, factorizations of the associated polynomials exist as well, so the same trick can be applied, though it is a bit complicated in performing the extended Euclidean algorithm.

It should be noted from Proposition 17 that the inverse of a rotational-add diffusion layer may not be a rotational-add one: additional scalars other than ± 1 may appear in the expression. For example, the inverse of $L_{n,m}^{\mathcal{G}}$ in Corollary 2 can be explicitly represented as

$$\begin{aligned}(L_{n,m}^{\mathcal{G}})^{-1}(\boldsymbol{x}) =& \frac{1}{15}\Bigg[-4\boldsymbol{x}-2\left(\boldsymbol{x}\lll\frac{m}{2}\right)+4\left(\boldsymbol{x}\lll m\right)+2\left(\boldsymbol{x}\lll\frac{3m}{2}\right)+\left(\boldsymbol{x}\lll 2m\right) \\ &+3\left(\boldsymbol{x}\lll\frac{5m}{2}\right)-\left(\boldsymbol{x}\lll 3m\right)+2\left(\boldsymbol{x}\lll\frac{7m}{2}\right)\Bigg],\end{aligned}$$

where $\boldsymbol{x}\in\mathbb{F}_p^{4m}$. Besides, representation of the inverse is probably somewhat complicated. However, this is a situation for many linear diffusion layers applied in symmetric ciphers.

To finish this part, we briefly discuss the implementation costs of inverses of the constructed rotational-add MDS diffusion layers. Firstly, it should be noted that with our constructing technique, the number of terms and the scalars in the inverse of a rotational-add MDS diffusion layer does not depend on the parameter m. For inverses of the three aforementioned classes of diffusion layers, they consist of 8, 12 and 12 terms, respectively, which implies that their implementations require 7, 11 and 11 rotations along with 7, 11 and 11 additions/subtractions of state vectors, respectively. The concrete complexity of scalar multiplications depends on the prime p. For example, $15^{-1}=1$ when $p=7$, bringing in no additional scalar multiplication, while $15^{-1}=30584$ when $p=65537$, resulting in a relatively big multiplication complexity. On the other hand, even though $(L_{4,m}^{\mathcal{G}})^{-1}$ demonstrates higher implementation cost compared to $L_{4,m}^{\mathcal{G}}$, this overhead may have low influence on efficiency of an advanced protocol such as hybrid FHE in practical applications. In such a setting, the symmetric decryption circuit is implemented homomorphically, in which scalar multiplications should be avoided as much as possible and thus $L_{4,m}^{\mathcal{G}}$ is used, while the symmetric encryption algorithm is implemented and performed in a conventional way. Due to the inherent noise accumulation and substantial computational overhead of current FHE schemes, greater emphasis should be put on optimizing decryption efficiency compared to encryption.

In Section 5.4, we will evaluate the efficiency of $L_{4,m}^{\mathcal{G}}$ and $(L_{4,m}^{\mathcal{G}})^{-1}$ in FHE settings by some computational experiments.

Table 1: Comparison of efficiency of YuX and YuX† decryption algorithms

Cipher	Parameters	p	nslots	Runtime (s)	Throughput (KB/min)
YuX	$(16 \times 16, 118, 9)$	65537	32768	65.39	939.59
YuX	$(16 \times 16, 128, 12)$	65537	32768	206.71	297.23
YuX	$(16 \times 16, 128, 14)$	65537	32768	311.74	197.09
YuX†	$\mathbf{(16 \times 16, 118, 9)}$	**65537**	**32768**	**63.25**	**971.38**
YuX†	$\mathbf{(16 \times 16, 128, 12)}$	**65537**	**32768**	**200.51**	**306.42**
YuX†	$\mathbf{(16 \times 16, 128, 14)}$	**65537**	**32768**	**298.93**	**205.53**

- Parameters (n, d, r): the bit-level blocksize, the data complexity and the number of rounds of the cipher;
- nslots: the number of slots in the homomorphic implementation, corresponding to the number of blocks processed in parallel with the state-sliced packed implementation proposed in [LLC$^+$24];
- Throughput: $=$ nslots $\cdot \frac{n \cdot 60 \cdot 2^{-13}}{\text{Runtime}}$.

5.4 The efficiency of proposed rotational-add diffusion layers

Taking the rotational-add MDS diffusion layer constructed in Corollary 2 as an example, we apply it to the YuX cipher by setting $m = 4$. The modified cipher is denoted as YuX†. The sole distinction between YuX and YuX† lies in their respective diffusion layers.

Recall that the associated polynomial of the diffusion layer in the YuX decryption algorithm is given by

$$1 + x^3 + x^4 + x^8 + x^9 + x^{12} + x^{14},$$

whose implementation needs 6 rotations and 6 additions, while the associated polynomial of the inverse diffusion layer in the YuX encryption algorithm is

$$\frac{1}{21} \cdot \big(12 + 5x + 5x^2 + 5x^3 + 12x^4 + 5x^5 + 5x^6 + 5x^7 - 9x^8 \\ - 16x^9 - 16x^{10} - 16x^{11} - 9x^{12} + 5x^{13} + 5x^{14} + 5x^{15}\big),$$

whose implementation needs 15 rotations, 10 additions and 5 subtractions.

We directly adopt the codes[2] for implementing YuX and modify them to obtain an impmementation of YuX†. Our experiments were conducted on an Ubuntu 22.04 LTS server equipped with 512GiB DDR4 ECC RAM and four AMD EPYC 7763 processors (64 cores/processor, 256 cores total) @ 2.45 GHz base frequency. The experimental evaluations are focused on two performance metrics: execution time and throughput, for homomorphically executed decryption circuits and conventional (non-homomorphic) encryption algorithms of YuX and YuX†. For the analysis of homomorphic decryption, in view of the expensive computational overhead of homomorphic operations, we only conducted 10 iterations of both ciphers to calculate the average runtime and throughput, respectively. The results are presented in Table 1. It turns out that YuX† admits an runtime reduction of approximately 3% and a throughput improvement of approximately 3% across all tested configurations compared to YuX. For plain encryption performances, we measured the runtime and throughput across 1000 iterations of both ciphers, with results presented in Table 2. YuX† shows a runtime reduction of approximately 18% and a throughput gain of approximately 22%. Therefore, our constructed diffusion layer demonstrates superior performance to YuX in both decryption and encryption operations, validating its theoretical advantage.

[2]See https://github.com/YuXenc/Yux_FHE_HElib

Table 2: Comparison of efficiency of YuX and YuX† encryption algorithms

Cipher	Parameters	p	Runtime ($\times 10^{-6}$ s)	Throughput (KB/min)
YuX	$(16 \times 16, 118, 9)$	65537	6.51	288018.43
YuX	$(16 \times 16, 128, 12)$	65537	8.77	213797.04
YuX	$(16 \times 16, 128, 14)$	65537	10.39	180461.98
YuX†	$\mathbf{(16 \times 16, 118, 9)}$	**65537**	**5.30**	**353773.58**
YuX†	$\mathbf{(16 \times 16, 128, 12)}$	**65537**	**7.17**	**261506.28**
YuX†	$\mathbf{(16 \times 16, 128, 14)}$	**65537**	**8.49**	**220848.06**

- Parameters (n, d, r): the bit-level blocksize, the data complexity and the number of rounds of the cipher;
- Throughput: $= \frac{n \cdot 60 \cdot 2^{-13}}{\text{Runtime}}$.

6 Conclusion

In this paper, we investigate characterizations and constructions of lightest rotational-add diffusion layers that are MDS over $(\mathbb{F}_p^m)^n$. We propose a theoretical framework for this class of diffusion layers and derive a lower bound for size of the rotation offsets to ensure the MDS property. When $n = 4$ and if this bound is attained, we can derive computationally tractable necessary and sufficient conditions in four cases. We further provide three explicit constructions along with their corresponding inverses for given parameters, whose MDS properties hold for most primes p. In aspect of designing AO symmetric ciphers, the proposed MDS diffusion layers provide new candidates and can offer advantages in such application scenarios as FHE. To the best of our knowledge, this work constitutes the first systematic framework for the study of rotational-add MDS diffusion layers and introduces novel constructing techniques that do not rely on auxiliary search procedures.

As part of future research, while this work establishes theoretical foundations for only two of the eight identified matrix types of $L_{4,m}^{\mathcal{G}}$, extending the analysis to the remaining six types needs to be considered. Besides, a full classification of rotational-add MDS diffusion layers $L_{4,m}^{\mathcal{G}}$ with $|\mathcal{G}| = 5$ is still unsolved now. The current work is restricted to the case $0 \in \mathcal{G}$. Furthermore, characterizations and constructions of lightest rotational-add MDS diffusion layers $L_{n,m}^{\mathcal{G}}$ for $n > 4$ may also be investigated with similar techniques exhibited in this paper. Crucially, inspired by this work, we should study other strategies to design lightweight diffusion layers for AO ciphers, but the primary problem that should be addressed is to define some metrics to evaluate implementation costs of diffusion layers over $\mathbb{F}_p$. Whereas over $\mathbb{F}_2$, the metrics of XOR-count, including d-XOR, s-XOR, g-XOR, etc., are usually used to measure the costs of diffusion layers in hardware implementations, these notions cannot directly be extended to $\mathbb{F}_p$.

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A Examples of rotational-add MDS matrices

We provide examples of $L^{\mathcal{G}}_{4,m}$ that are MDS in the case $\mathcal{G} = \{0, g_2, g_2+m, g_2+2m, 3m\}$ for the parameters $p = 65537$ and $m = 4,\ 8$, for all the 16 cases in Figure 1. These examples were obtained by an exhaustive search of g_2 in each case. We list only the set $\mathcal{I}$ and $\mathcal{J}$ (see Definition 3) for each result.

Table 3: Searching results of 16 types of rotational-add MDS matrices with $m = 4$

Type	g_2	Set $\mathcal{I}$	Set $\mathcal{J}$
Case 1	1	{0, 1, 5, 9, 12}	{}
	3	{0, 3, 7, 11, 12}	{}
Case 2	1	{1, 5, 9, 12}	{0}
	2	{2, 6, 10, 12}	{0}
	3	{3, 7, 11, 12}	{0}
Case 3	1	{0, 5, 9, 12}	{1}
	3	{0, 7, 11, 12}	{3}
Case 4	1	{0, 1, 9, 12}	{5}
	3	{0, 3, 11, 12}	{7}
Case 5	1	{0, 1, 5, 12}	{9}
	3	{0, 3, 7, 12}	{11}
Case 6	1	{0, 1, 5, 9}	{12}
	2	{0, 2, 6, 10}	{12}
	3	{0, 3, 7, 11}	{12}
Case 7	1	{5, 9, 12}	{0, 1}
	3	{7, 11, 12}	{0, 3}
Case 8	1	{1, 9, 12}	{0, 5}
	3	{3, 11, 12}	{0, 7}
Case 9	1	{1, 5, 12}	{0, 9}
	3	{3, 7, 12}	{0, 11}
Case 10	1	{1, 5, 9}	{0, 12}
	3	{3, 7, 11}	{0, 12}
Case 11	1	{0, 9, 12}	{1, 5}
	3	{0, 11, 12}	{3, 7}
Case 12	1	{0, 5, 12}	{1, 9}
	3	{0, 7, 12}	{3, 11}
Case 13	1	{0, 5, 9}	{1, 12}
	3	{0, 7, 11}	{3, 12}

Case 14	1	{0, 1, 12}	{5, 9}
	3	{0, 3, 12}	{7, 11}
Case 15	1	{0, 1, 9}	{5, 12}
	3	{0, 3, 11}	{7, 12}
Case 16	1	{0, 1, 5}	{9, 12}
	3	{0, 3, 7}	{11, 12}

Table 4: Searching results of 16 types of rotational-add MDS matrices with $m = 8$

Type	g_2	Set $\mathcal{I}$	Set $\mathcal{J}$
Case 1	2	{0, 2, 10, 18, 24}	{}
	3	{0, 3, 11, 19, 24}	{}
	5	{0, 5, 13, 21, 24}	{}
	6	{0, 6, 14, 22, 24}	{}
Case 2	1	{1, 9, 17, 24}	{0}
	2	{2, 10, 18, 24}	{0}
	3	{3, 11, 19, 24}	{0}
	4	{4, 12, 20, 24}	{0}
	5	{5, 13, 21, 24}	{0}
	6	{6, 14, 22, 24}	{0}
	7	{7, 15, 23, 24}	{0}
Case 3	1	{0, 9, 17, 24}	{1}
	2	{0, 10, 18, 24}	{2}
	3	{0, 11, 19, 24}	{3}
	5	{0, 13, 21, 24}	{5}
	6	{0, 14, 22, 24}	{6}
Case 4	2	{0, 2, 18, 24}	{10}
	3	{0, 3, 19, 24}	{11}
	5	{0, 5, 21, 24}	{13}
	6	{0, 6, 22, 24}	{14}
Case 5	2	{0, 2, 10, 24}	{18}
	3	{0, 3, 11, 24}	{19}
	5	{0, 5, 13, 24}	{21}
	6	{0, 6, 14, 24}	{22}
	7	{0, 7, 15, 24}	{23}
Case 6	1	{0, 1, 9, 17}	{24}
	2	{0, 2, 10, 18}	{24}
	3	{0, 3, 11, 19}	{24}
	4	{0, 4, 12, 20}	{24}
	5	{0, 5, 13, 21}	{24}
	6	{0, 6, 14, 22}	{24}
	7	{0, 7, 15, 23}	{24}
Case 7	2	{10, 18, 24}	{0, 2}
	3	{11, 19, 24}	{0, 3}
	5	{13, 21, 24}	{0, 5}
	6	{14, 22, 24}	{0, 6}
Case 8	2	{2, 18, 24}	{0, 10}
	3	{3, 19, 24}	{0, 11}
	5	{5, 21, 24}	{0, 13}
	6	{6, 22, 24}	{0, 14}
Case 9	2	{2, 10, 24}	{0, 18}

	3	{3, 11, 24}	{0, 19}
	5	{5, 13, 24}	{0, 21}
	6	{6, 14, 24}	{0, 22}
Case 10	2	{2, 10, 18}	{0, 24}
	3	{3, 11, 19}	{0, 24}
	5	{5, 13, 21}	{0, 24}
	6	{6, 14, 22}	{0, 24}
Case 11	2	{0, 18, 24}	{2, 10}
	3	{0, 19, 24}	{3, 11}
	5	{0, 21, 24}	{5, 13}
	6	{0, 22, 24}	{6, 14}
	7	{0, 23, 24}	{7, 15}
Case 12	2	{0, 10, 24}	{2, 18}
	3	{0, 11, 24}	{3, 19}
	5	{0, 13, 24}	{5, 21}
	6	{0, 14, 24}	{6, 22}
Case 13	2	{0, 10, 18}	{2, 24}
	3	{0, 11, 19}	{3, 24}
	5	{0, 13, 21}	{5, 24}
	6	{0, 14, 22}	{6, 24}
Case 14	1	{0, 1, 24}	{9, 17}
	2	{0, 2, 24}	{10, 18}
	3	{0, 3, 24}	{11, 19}
	5	{0, 5, 24}	{13, 21}
	6	{0, 6, 24}	{14, 22}
Case 15	2	{0, 2, 18}	{10, 24}
	3	{0, 3, 19}	{11, 24}
	5	{0, 5, 21}	{13, 24}
	6	{0, 6, 22}	{14, 24}
Case 16	2	{0, 2, 10}	{18, 24}
	3	{0, 3, 11}	{19, 24}
	5	{0, 5, 13}	{21, 24}
	6	{0, 6, 14}	{22, 24}

B More constructions of rotational-add MDS diffusion layers with various m

Table 5, Table 6 and Table 7 present constructions of $L^{\mathcal{G}}_{4,m}$ that are MDS corresponding to Case 2, Case 4 and Case 12 in Figure 1, respectively. In each case, we traverse the values of an integer t from 2 up to 16 under the condition $t \mid m$ for any m, listing all possible g_2 and the corresponding conditions for $L^{\mathcal{G}}_{4,m}$ to be MDS in terms of excluded primes p. These results can be easily obtained from Theorem 5 and Theorem 6 through simple SageMath codes. Note that in these tables, an entry "–" represents non-existence of MDS matrices for the corresponding parameters. Besides, for Case 4 and Case 12 (in Figure 1), we need only to consider values of g_2 satisfying $g_2 > m/2$ as discussed in Section 5.2.

Table 5: Construction results for Case 2

m	g_2	Excluded p
$2 \mid m$	$m/2$	[2, 3, 5]

$3 \mid m$	$m/3$	[2, 3, 11, 13, 47]
	$2m/3$	[2, 5, 7, 11, 61, 101]
$4 \mid m$	$m/4$	[7, 11, 13, 17, 29, 61, 521, 641]
	$3m/4$	[7, 11, 13, 17, 29, 61, 521, 641]
$5 \mid m$	$m/5$	–
	$2m/5$	[3, 5, 7, 29, 31, 37, 41, 47, 67, 2591]
	$3m/5$	[3, 7, 11, 19, 23, 29, 41, 43, 311]
	$4m/5$	[2, 17, 19, 23, 31, 41, 53, 131]
$6 \mid m$	$m/6$	[5, 7, 13, 17, 29, 31, 199, 449, 7927, 17203]
	$5m/6$	[5, 7, 13, 17, 29, 31, 199, 449, 7927, 17203]
$7 \mid m$	$m/7$	[3, 7, 43, 113, 457, 521, 10181, 10789]
	$2m/7$	[2, 5, 19, 29, 31, 67, 127, 613, 2383, 18911]
	$3m/7$	[3, 5, 17, 29, 31, 37, 43, 53, 199, 20389]
	$4m/7$	[7, 11, 17, 31, 73, 113, 127, 131, 199, 827, 47581]
	$5m/7$	–
	$6m/7$	[3, 5, 13, 19, 29, 37, 127, 277, 337, 521, 1483, 1709]
$8 \mid m$	$m/8$	[2, 11, 19, 31, 61, 103, 127, 257, 1181, 4357, 33851]
	$3m/8$	[2, 7, 13, 19, 29, 31, 73, 199, 257, 521, 16829, 215909]
	$5m/8$	[2, 7, 13, 19, 29, 31, 73, 199, 257, 521, 16829, 215909]
	$7m/8$	[2, 11, 19, 31, 61, 103, 127, 257, 1181, 4357, 33851]
$9 \mid m$	$m/9$	[3, 5, 17, 19, 23, 61, 109, 3571, 4651, 14071, 191899]
	$2m/9$	[3, 5, 7, 47, 73, 109, 127, 131, 139, 199, 726647, 2093881]
	$4m/9$	[7, 13, 17, 23, 29, 31, 37, 73, 521, 613, 1483, 2083, 80671]
	$5m/9$	[3, 5, 11, 13, 19, 109, 239, 457, 521, 374069, 457517]
	$7m/9$	[3, 13, 19, 37, 43, 47, 53, 199, 4621, 694259]
	$8m/9$	[7, 11, 13, 37, 67, 73, 257, 733, 1301, 3517, 3571, 609113]
$10 \mid m$	$m/10$	[2, 3, 5, 7, 11, 41, 73, 199, 617, 9349, 274361, 1511269]
	$3m/10$	[5, 7, 17, 31, 41, 127, 449, 521, 577, 709, 3571, 12577, 3888067]
	$7m/10$	[5, 7, 17, 31, 41, 127, 449, 521, 577, 709, 3571, 12577, 3888067]
	$9m/10$	[2, 3, 5, 7, 11, 41, 73, 199, 617, 9349, 274361, 1511269]
$11 \mid m$	$m/11$	–
	$2m/11$	[5, 7, 11, 23, 31, 47, 59, 73, 89, 397, 521, 65657, 228013, 1237853]
	$3m/11$	[3, 5, 13, 17, 19, 239, 379, 397, 683, 9349, 826663, 1839059]
	$4m/11$	[2, 11, 17, 19, 23, 31, 89, 127, 1301, 1483, 2113, 316429, 9566093]
	$5m/11$	[2, 3, 7, 11, 43, 47, 239, 683, 2113, 3571, 4799, 8388691]
	$6m/11$	[2, 3, 5, 7, 13, 23, 31, 41, 47, 89, 139, 379, 397, 3571, 611033, 30989039]

	$7m/11$	–
	$8m/11$	[7, 23, 29, 37, 89, 257, 463, 571, 613, 811, 2113, 9349, 28793, 101917]
	$9m/11$	[3, 19, 31, 41, 521, 683, 2113, 21577, 4429769, 6463643]
	$10m/11$	[2, 5, 23, 29, 41, 89, 131, 211, 397, 1117, 2761999, 4484789]
$12 \mid m$	$m/12$	[2, 17, 23, 29, 89, 139, 241, 461, 521, 857, 10099, 37517, 53407, 442069]
	$5m/12$	[17, 31, 41, 127, 197, 241, 3571, 4357, 9349, 20231, 20593, 107632313]
	$7m/12$	[17, 31, 41, 127, 197, 241, 3571, 4357, 9349, 20231, 20593, 107632313]
	$11m/12$	[2, 17, 23, 29, 89, 139, 241, 461, 521, 857, 10099, 37517, 53407, 442069]
$13 \mid m$	$m/13$	[3, 5, 7, 11, 13, 17, 31, 47, 101, 149, 151, 1613, 2731, 3109, 5323, 13913, 82139]
	$2m/13$	[2, 3, 5, 11, 23, 31, 89, 613, 1069, 1613, 7351, 8191, 378977, 15377827]
	$3m/13$	[3, 11, 23, 31, 47, 53, 139, 157, 457, 461, 1277, 1453, 1663, 2731, 8747, 263303]
	$4m/13$	[2, 5, 7, 17, 53, 73, 157, 199, 379, 1117, 3571, 8191, 62641577, 371930747]
	$5m/13$	–
	$6m/13$	[5, 11, 13, 19, 23, 37, 127, 811, 1301, 1613, 8191, 9349, 1601023, 566670439]
	$7m/13$	[3, 7, 41, 43, 53, 157, 233, 317, 379, 2731, 4651, 9349, 198823, 1000457]
	$8m/13$	[2, 19, 29, 31, 47, 53, 139, 157, 211, 257, 8191, 65657, 13489271, 114713371]
	$9m/13$	[2, 3, 5, 19, 43, 89, 1613, 2731, 3571, 46811, 118280137, 124338157]
	$10m/13$	[3, 5, 7, 37, 41, 67, 139, 461, 1483, 1613, 8191, 58763, 1313567263]
	$11m/13$	–
	$12m/13$	[11, 17, 29, 31, 43, 47, 53, 101, 151, 157, 241, 8191, 1385429, 4646533, 42539671]
$14 \mid m$	$m/14$	[2, 5, 11, 17, 19, 29, 31, 37, 107, 113, 127, 401, 449, 641, 5779, 8191, 5713759, 24657623]
	$3m/14$	[3, 5, 7, 11, 19, 23, 29, 43, 89, 101, 107, 113, 151, 191, 241, 1297, 3571, 11173, 1339127]
	$5m/14$	[5, 7, 17, 29, 31, 73, 113, 127, 139, 461, 577, 4259, 7433, 9349, 204353, 264743, 1069307]
	$9m/14$	[5, 7, 17, 29, 31, 73, 113, 127, 139, 461, 577, 4259, 7433, 9349, 204353, 264743, 1069307]
	$11m/14$	[3, 5, 7, 11, 19, 23, 29, 43, 89, 101, 107, 113, 151, 191, 241, 1297, 3571, 11173, 1339127]

	$13m/14$	[2, 5, 11, 17, 19, 29, 31, 37, 107, 113, 127, 401, 449, 641, 5779, 8191, 5713759, 24657623]
$15 \mid m$	$m/15$	[3, 11, 13, 41, 43, 47, 59, 61, 127, 173, 229, 239, 331, 1933, 2731, 2801, 19489, 26489, 6628679]
	$2m/15$	[5, 7, 13, 29, 31, 83, 109, 151, 157, 1321, 1483, 3571, 8191, 77081, 262351, 45146872709]
	$4m/15$	[7, 13, 17, 23, 31, 41, 47, 61, 89, 139, 151, 521, 577, 9349, 129379, 229249, 1385429, 11815043]
	$7m/15$	[3, 5, 11, 17, 43, 89, 107, 139, 331, 461, 1321, 5927, 21577, 46811, 888163, 39884269]
	$8m/15$	[5, 7, 13, 31, 41, 59, 61, 127, 139, 151, 199, 257, 461, 1117, 65657, 267724777, 12800504581]
	$11m/15$	[3, 5, 11, 17, 47, 149, 211, 233, 331, 683, 1321, 3109, 9349, 32533, 2756339]
	$13m/15$	[3, 11, 13, 41, 61, 281, 331, 457, 2731, 3571, 2135117, 36319529, 15049391911]
	$14m/15$	[3, 5, 7, 23, 29, 31, 41, 59, 113, 151, 239, 431, 613, 1321, 2749, 19489, 21487, 31907, 15201799]
$16 \mid m$	$m/16$	[7, 13, 19, 31, 73, 151, 277, 3571, 5717, 6803, 7559, 65537, 156217, 3010349, 817674491]
	$3m/16$	[2, 7, 13, 29, 47, 59, 3109, 4357, 8191, 9349, 19489, 65537, 4109473279, 23610122867]
	$5m/16$	[2, 19, 23, 29, 31, 89, 107, 211, 223, 1109, 5779, 20593, 65537, 1329457, 2033989, 8112787]
	$7m/16$	[7, 11, 13, 37, 61, 73, 101, 127, 139, 151, 239, 461, 2351, 65537, 442069, 9426181, 23297683]
	$9m/16$	[7, 11, 13, 37, 61, 73, 101, 127, 139, 151, 239, 461, 2351, 65537, 442069, 9426181, 23297683]
	$11m/16$	[2, 19, 23, 29, 31, 89, 107, 211, 223, 1109, 5779, 20593, 65537, 1329457, 2033989, 8112787]
	$13m/16$	[2, 7, 13, 29, 47, 59, 3109, 4357, 8191, 9349, 19489, 65537, 4109473279, 23610122867]
	$15m/16$	[7, 13, 19, 31, 73, 151, 277, 3571, 5717, 6803, 7559, 65537, 156217, 3010349, 817674491]

Table 6: Construction results for Case 4

m	g_2	Excluded p
$3 \mid m$	$2m/3$	[2, 3, 11, 13, 47]
$4 \mid m$	$3m/4$	[3, 5, 11, 17, 23, 29, 107]
$5 \mid m$	$3m/5$	[3, 7, 11, 19, 23, 29, 41, 43, 311]
	$4m/5$	–

$6 \mid m$	$5m/6$	[3, 7, 11, 17, 29, 199, 269, 661]
$7 \mid m$	$4m/7$	[3, 5, 17, 29, 31, 37, 43, 53, 199, 20389]
	$5m/7$	–
	$6m/7$	[3, 7, 43, 113, 457, 521, 10181, 10789]
$8 \mid m$	$5m/8$	[2, 3, 5, 11, 17, 37, 41, 179, 199, 257, 521, 557, 599]
	$7m/8$	–
$9 \mid m$	$5m/9$	[3, 5, 11, 13, 19, 109, 239, 457, 521, 374069, 457517]
	$7m/9$	[3, 13, 19, 37, 43, 47, 53, 199, 4621, 694259]
	$8m/9$	[3, 5, 17, 19, 23, 61, 109, 3571, 4651, 14071, 191899]
$10 \mid m$	$7m/10$	[3, 11, 13, 17, 31, 43, 521, 3571, 15737, 331511]
	$9m/10$	[2, 3, 11, 13, 19, 29, 31, 37, 199, 9349, 40829, 866083]
$11 \mid m$	$6m/11$	[2, 3, 7, 11, 43, 47, 239, 683, 2113, 3571, 4799, 8388691]
	$7m/11$	–
	$8m/11$	[3, 5, 13, 17, 19, 239, 379, 397, 683, 9349, 826663, 1839059]
	$9m/11$	[3, 19, 31, 41, 521, 683, 2113, 21577, 4429769, 6463643]
	$10m/11$	–
$12 \mid m$	$7m/12$	[3, 5, 7, 11, 13, 17, 23, 43, 53, 67, 131, 139, 151, 241, 3571, 9349, 236653, 392339]
	$11m/12$	[2, 3, 5, 7, 13, 17, 139, 239, 241, 461, 521, 613, 683, 3539, 4423, 9857, 12809]
$13 \mid m$	$7m/13$	[3, 7, 41, 43, 53, 157, 233, 317, 379, 2731, 4651, 9349, 198823, 1000457]
	$8m/13$	–
	$9m/13$	[2, 3, 5, 19, 43, 89, 1613, 2731, 3571, 46811, 118280137, 124338157]
	$10m/13$	[3, 11, 23, 31, 47, 53, 139, 157, 457, 461, 1277, 1453, 1663, 2731, 8747, 263303]
	$11m/13$	–
	$12m/13$	[3, 5, 7, 11, 13, 17, 31, 47, 101, 149, 151, 1613, 2731, 3109, 5323, 13913, 82139]
$14 \mid m$	$9m/14$	[3, 11, 13, 17, 19, 23, 37, 43, 127, 139, 449, 461, 557, 2729, 9349, 3162659]
	$11m/14$	[3, 11, 13, 37, 43, 61, 73, 101, 127, 151, 307, 677, 683, 2389, 3571, 11119, 60161]
	$13m/14$	–
$15 \mid m$	$8m/15$	[3, 5, 11, 17, 43, 89, 107, 139, 331, 461, 1321, 5927, 21577, 46811, 888163, 39884269]
	$11m/15$	[3, 5, 11, 17, 47, 149, 211, 233, 331, 683, 1321, 3109, 9349, 32533, 2756339]
	$13m/15$	[3, 11, 13, 41, 61, 281, 331, 457, 2731, 3571, 2135117, 36319529, 15049391911]

	$14m/15$	[3, 11, 13, 41, 43, 47, 59, 61, 127, 173, 229, 239, 331, 1933, 2731, 2801, 19489, 26489, 6628679]
$16 \mid m$	$9m/16$	[3, 5, 11, 17, 19, 31, 43, 101, 139, 151, 239, 257, 461, 613, 937, 5479, 12163, 65537, 19480367]
	$11m/16$	–
	$13m/16$	[2, 3, 5, 17, 23, 43, 47, 59, 67, 257, 379, 571, 2731, 9349, 19489, 65537, 2072039, 112749881]
	$15m/16$	[3, 5, 11, 17, 23, 37, 41, 47, 139, 151, 257, 331, 2129, 3571, 65537, 101611, 3010349, 5486861]

Table 7: Construction results for Case 12

m	g_2	excluded p
$3 \mid m$	$2m/3$	[2, 3, 5, 7, 11, 61, 101]
$4 \mid m$	$3m/4$	[3, 5, 11, 17, 23, 29, 107]
$5 \mid m$	$3m/5$	[3, 5, 7, 29, 31, 37, 41, 47, 67, 2591]
	$4m/5$	[2, 3, 17, 19, 23, 31, 41, 53, 131]
$6 \mid m$	$5m/6$	[3, 7, 11, 17, 29, 199, 269, 661]
$7 \mid m$	$4m/7$	[3, 7, 11, 17, 31, 73, 113, 127, 131, 199, 827, 47581]
	$5m/7$	[2, 3, 5, 19, 29, 31, 67, 127, 613, 2383, 18911]
	$6m/7$	[3, 5, 13, 19, 29, 37, 127, 277, 337, 521, 1483, 1709]
$8 \mid m$	$5m/8$	[2, 3, 5, 11, 17, 37, 41, 179, 199, 257, 521, 557, 599]
	$7m/8$	–
$9 \mid m$	$5m/9$	[3, 7, 13, 17, 23, 29, 31, 37, 73, 521, 613, 1483, 2083, 80671]
	$7m/9$	[3, 5, 7, 47, 73, 109, 127, 131, 139, 199, 726647, 2093881]
	$8m/9$	[3, 7, 11, 13, 37, 67, 73, 257, 733, 1301, 3517, 3571, 609113]
$10 \mid m$	$7m/10$	[3, 11, 13, 17, 31, 43, 521, 3571, 15737, 331511]
	$9m/10$	[2, 3, 11, 13, 19, 29, 31, 37, 199, 9349, 40829, 866083]
$11 \mid m$	$6m/11$	[2, 3, 5, 7, 13, 23, 31, 41, 47, 89, 139, 379, 397, 3571, 611033, 30989039]
	$7m/11$	[2, 3, 11, 17, 19, 23, 31, 89, 127, 1301, 1483, 2113, 316429, 9566093]
	$8m/11$	[3, 7, 23, 29, 37, 89, 257, 463, 571, 613, 811, 2113, 9349, 28793, 101917]
	$9m/11$	[3, 5, 7, 11, 23, 31, 47, 59, 73, 89, 397, 521, 65657, 228013, 1237853]
	$10m/11$	[2, 3, 5, 23, 29, 41, 89, 131, 211, 397, 1117, 2761999, 4484789]
$12 \mid m$	$7m/12$	[3, 5, 7, 11, 13, 17, 23, 43, 53, 67, 131, 139, 151, 241, 3571, 9349, 236653, 392339]

	$11m/12$	[2, 3, 5, 7, 13, 17, 139, 239, 241, 461, 521, 613, 683, 3539, 4423, 9857, 12809]
$13 \mid m$	$7m/13$	[3, 5, 11, 13, 19, 23, 37, 127, 811, 1301, 1613, 8191, 9349, 1601023, 566670439]
	$8m/13$	[2, 3, 19, 29, 31, 47, 53, 139, 157, 211, 257, 8191, 65657, 13489271, 114713371]
	$9m/13$	[2, 3, 5, 7, 17, 53, 73, 157, 199, 379, 1117, 3571, 8191, 62641577, 371930747]
	$10m/13$	[3, 5, 7, 37, 41, 67, 139, 461, 1483, 1613, 8191, 58763, 1313567263]
	$11m/13$	[2, 3, 5, 11, 23, 31, 89, 613, 1069, 1613, 7351, 8191, 378977, 15377827]
	$12m/13$	[3, 11, 17, 29, 31, 43, 47, 53, 101, 151, 157, 241, 8191, 1385429, 4646533, 42539671]
$14 \mid m$	$9m/14$	[3, 11, 13, 17, 19, 23, 37, 43, 127, 139, 449, 461, 557, 2729, 9349, 3162659]
	$11m/14$	[3, 11, 13, 37, 43, 61, 73, 101, 127, 151, 307, 677, 683, 2389, 3571, 11119, 60161]
	$13m/14$	–
$15 \mid m$	$8m/15$	[3, 5, 7, 13, 31, 41, 59, 61, 127, 139, 151, 199, 257, 461, 1117, 65657, 267724777, 12800504581]
	$11m/15$	[3, 7, 13, 17, 23, 31, 41, 47, 61, 89, 139, 151, 521, 577, 9349, 129379, 229249, 1385429, 11815043]
	$13m/15$	[3, 5, 7, 13, 29, 31, 83, 109, 151, 157, 1321, 1483, 3571, 8191, 77081, 262351, 45146872709]
	$14m/15$	[3, 5, 7, 23, 29, 31, 41, 59, 113, 151, 239, 431, 613, 1321, 2749, 19489, 21487, 31907, 15201799]
$16 \mid m$	$9m/16$	[3, 5, 11, 17, 19, 31, 43, 101, 139, 151, 239, 257, 461, 613, 937, 5479, 12163, 65537, 19480367]
	$11m/16$	–
	$13m/16$	[2, 3, 5, 17, 23, 43, 47, 59, 67, 257, 379, 571, 2731, 9349, 19489, 65537, 2072039, 112749881]
	$15m/16$	[3, 5, 11, 17, 23, 37, 41, 47, 139, 151, 257, 331, 2129, 3571, 65537, 101611, 3010349, 5486861]